



HOME SERVICES,
SIMPLIFIED.

ServiGo

PROJECT DOCUMENTATION

An end-to-end home service booking platform that connects customers with skilled technicians through secure payments, real-time tracking, and a modern mobile experience.

01

MOBILE APP

Customers & technicians

02

ADMIN PANEL

Laravel management portal

03

REAL-TIME

Location & notifications

Android / Java

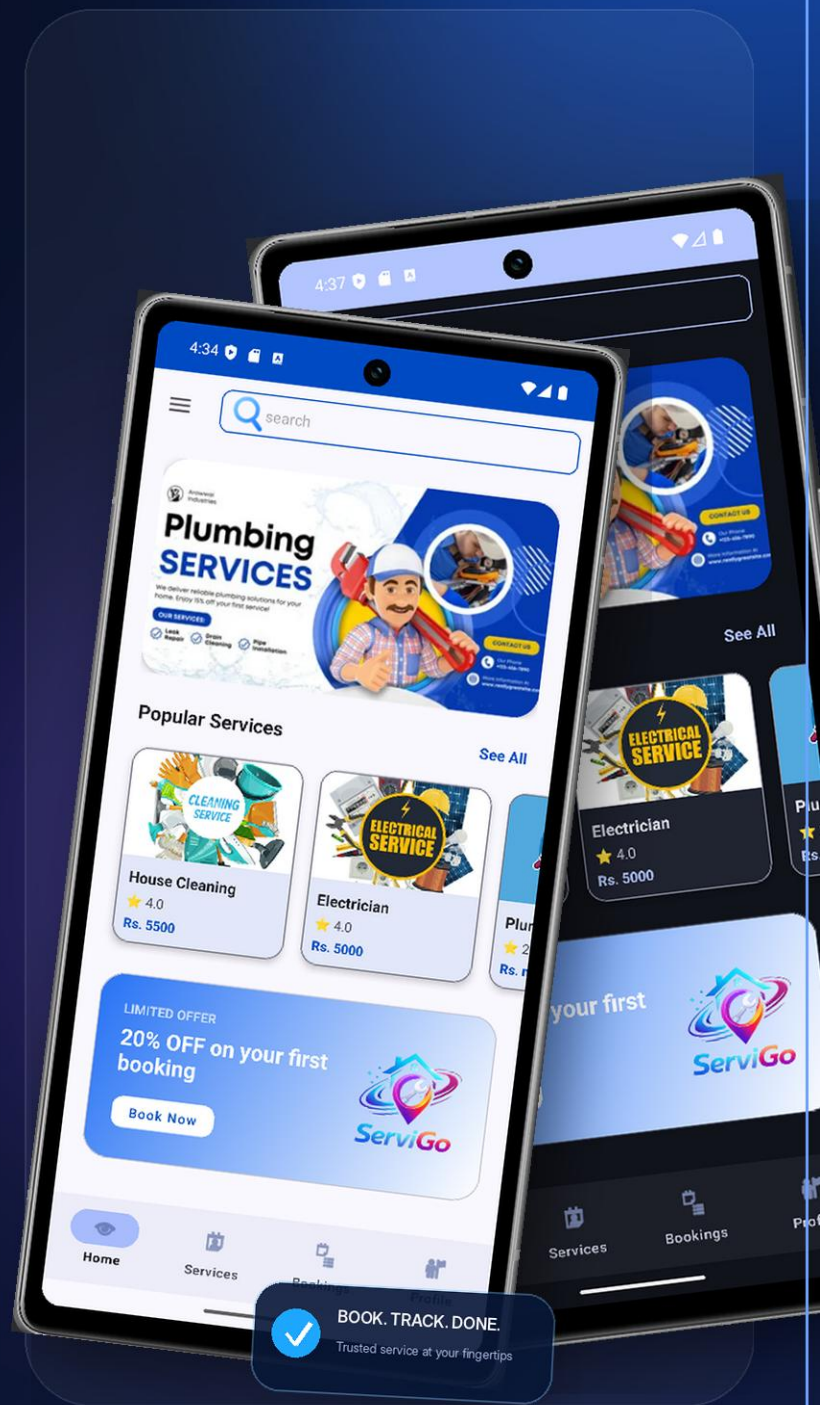
Laravel / PHP

Firebase

MySQL

Google Maps

PayHere



PREPARED BY

Tevinu Manuditha Wijesinghe

Software Engineering Project

PLATFORM

Android Mobile App + Web Admin

REST API • Cloud Services • Secure Payments

i. Acknowledgement

I would like to express my gratitude to all those who supported me to build this ServiGo app successfully and who guided me throughout development. I would like to specially thanks to my lecturers for their greatest guidance and knowledge. Also my friends who supported me and my family for their encouragement. Their support made it possible to complete ServiGo app successfully.

ii. Abstract

ServiGo is a mobile app that developed to connect home service seekers and skilled technicians. This system is provide a solution as end to end with search services, booking service, secure online payments and real time tracking.

This project included two platforms. An android mobile app for customers and technicians and a website for administrators. The android mobile app is developed using java and fragment architecture. Backend is developed using PHP and RESTful APIs with MySQL. This project uses firebase authentication, firebase firestore, firebase storage for store user profile images, real time data updating and push notifications

ServiGo represent a modern mobile application and website that uses modern technologies to ensure reliable, efficient and user friendly experience to users.

iii. Table of Content

1. Introduction	9
1.1. Project Overview	9
1.2. Objectives	9
2. Background	10
3. Specifications	10
3.1. User Roles	10
3.2. Features and Functionalities	11
3.3. Functional Requirement	13
3.4. Non functional Requirements	13
4. Design	14
4.1. System Architecture	14
4.2. Technology Stack	15
4.3. Project Architecture	15
4.4. Use Case Diagram	16
4.5. Class Diagram	18
4.6. Sequence Diagram	19
4.7. ER Diagram (MySQL)	20
4.8. ER Diagram (Firebase Firestore)	21
4.9. Combined ER Diagram (MySQL + Firebase Firestore)	22
4.10. User Interface (UI) Design	23
5. Implementaion	59
5.1. Backend Implementation	59
5.2. Frontend Implementation	59
5.3. Database Design	59
5.4. API Endpoints (PHP Backend)	60
5.5. Security Measures	60
5.6. Deployment	60
6. Result and Evaluation	61
6.1. System Functionality testing	61

6.2. System performance	63
6.3. User Experience Evaluation	63
6.4. Limitaions	64
6.5. Overall Evaluation	64
7. Future Work	64
8. Conclusion	64
9. Reflection	65
10. References	66

iv. Table of Figures

Figure 1 : System Architecture	14
Figure 2 : Use Case Diagram	17
Figure 3 : Class Diagram	18
Figure 4 : Sequence Diagram	19
Figure 5 : MySQL ER Diagram	20
Figure 6 :ER Diagram (Firebase)	21
Figure 7 : Combined ER Diagram (MySQL + Firebase)	22
Figure 8 : Splash Scree Light Theme	23
Figure 9 : Splash Screen Dark Theme	24
Figure 10 : Sign in Light theme	25
Figure 11 : Sign In Dark Theme	26
Figure 12 : Sign Up Screen Light	27
Figure 13 : Sign Up Screen Dark	28
Figure 14 : Customer Home screen Light theme	29
Figure 15 : Customer Home screen Dark theme	30
Figure 16 : Side navigation bar Light Theme	31
Figure 17 : Side navigation bar DarkTheme	32
Figure 18 : Services Screen Light Theme	33
Figure 19 : Services Screen Dark Theme	34
Figure 20 : Single service screen light theme	35
Figure 21 :Single service screen light theme	36
Figure 22 :Booking From	37
Figure 23 : Payment Phase	39
Figure 24 Payment confirmation	41
Figure 25 : Profile Screen	43
Figure 26 : Booking History	45
Figure 27 : Map	47
Figure 28 : Review Section	49
Figure 29 : Complete Technician Profile Screen Light Theme	51

Figure 30 : Complete Technician Profile Screen Dark Theme.	52
Figure 31 : Technician Home	53
Figure 32 : Technician booking history	54
Figure 33 : Status Changing	55
Figure 34 : Admin Login	56
Figure 35 : Booking management(Dashboard)	57
Figure 36 : Technician Assignment	57
Figure 37 : User management	58
Figure 38 : Payment approved	62
Figure 39 : Push Notification	63

v. List of Tables

Table 1 : Technology Stack	15
Table 2 : Project Architecture	16
Table 3 : API Endpoints	60

1. Introduction

1.1. Project Overview

ServiGo is a modern service booking platform (M-Commerce) that developed to connect home service seekers and well skilled technicians. It is a very efficient and reliable platform. ServiGo project is a dual interface system.

1. Android mobile application - This application for the customers to book services and for technicians to manage assignments.
2. Laravel web admin panel - This application is for admin to handle bookings, oversee the system and manage users.

The goal of the ServiGo system is to simplify day to day tasks by providing end to end experience for users from browsing services, scheduling appointments, making secure payments and real time location tracking on google map.

1.2. Objectives

- To simplify service selection and booking services through user friendly modern platform.

ServiGo is developed to find services without any confusion. The interface of ServiGo is very simple and user friendly, it allows customers to choose the service easily and schedule appointment in just few steps.

- To provide a seamless and real time booking experience.

The ServiGo allows users to book services smoothly without any delays and without confusions. With real time updates, user able to see available options instantly. That ensure efficiency and reliability of booking process.

- To enable real time technician tracking.

ServiGo allows customers to track technician live location smoothly on google map after technician assigned. This ensure guarantee of the system and it helps customer to know exactly when technician will arrive.

- To provide secure, non confusion and reliable cashless payment.

The ServiGo platform implements a secure and trusted payment gateway to payment process. This allows the customers to make checkout experience safely and smoothly without cash and any issue. Also ensures transparency and reduces risks with physical transaction.

- To enhance customer trust about the ServiGo services through the rating and reviews of the services.

In ServiGo app, customer able to see others reviews and ratings for the services and also they can leave their feedback after service is completed. This ensure the service quality and build the trust in platform.

- To Control full system by administrator.

The admin panel of ServiGo is allows administrators to manage user accounts and manage bookings, assign technicians, monitor system and handle failures. This ensures the ServiGo system is runs smoothly and efficiently without any issues and it helps to identify issues.

- Collect data to analyze and get decisions.

The ServiGo system collects data and organize the data related to bookings, users and services. This helps administrators and developers to improve the ServiGo system in future.

2. Background

In present, world is fast and people growingly rely on digital platforms to simplify their day to day tasks. But finding a professional skilled, reliable providers for the tasks such as painting, plumbing, carpentering, cleaning are very difficult now. Traditionally searching service providers by personal recommendations or by advertisement. It is leads to time wasting, unreliable and not guaranteed.

Many service platforms are providing simple solutions but not the complete solution. Some platforms runs without any verification of technicians, wasting users time, do not provide any guarantee. It may leads to fail the whole system. Some users face to difficulties with scheduling services, tracking technician location, trust, quality and efficiency.

In today's fast growing of the mobile technology and cloud based services gives opportunity to improve the process of how the service providing platforms are runs. The real time database, real time tracking and mobile payments can implement to service providing platforms to make more efficient, reliable and more user friendly experience to users.

ServiGo is developed to directly deal with these challenges by providing trust full, reliable guaranteed platform that connects customers and skilled technicians. ServiGo developed by using modern technologies such as cloud services, real time tracking to provide smooth experience. The features of ServiGo like review system, live technician tracking and secure payments without physical cash leads to improve efficiency, reliability, transparency and user friendly experience.

3. Specifications

3.1. User Roles

Customer

- Customer able to sign in and sign up.
- Search services from home page.
- View booking history.
- Customer can refresh the booking history by shaking the phone.
- Add profile image, update thier profile details.
- Booking a service by set date, time, address, mobile, email address and name
- Make secure payments with payhere.

- Track technician location in real time.
- Customer can call to the assigned technician.
- Leave review and submit rating after service is completed.
- Customer able to receive emails and notifications.
- Customer able to log out.

Technician

- Customer able to sign in and sign up.
- Add profile image, update profile details.
- View assigned bookings.
- Technician can refresh the booking history by shaking the phone.
- Technician can call to the customer
- View booking history.
- View customer location on google map.
- Change the status of the booking after arrived and completed.
- Share real time location with customers.
- Technician able to Log out from ServiGo.

Admin

- Log in to the admin panel.
- Manage users (ban / unban).
- Oversees users, bookings.
- Update status of bookings.
- Assign technicians to the bookings
- Monitor system and analyze.
- Delete bookings.

3.2. Features and Functionalities

Mobile Application

- **Authentication**

The customers and technicians are allowed to log in to the ServiGo by using firebase authentication.

- **Find services**

Customer allows to find services from search bar, services category fragment and home fragment. Customer able to view details by clicking a services and in that space show ratings and reviews also.

- **Booking System**

Customers can booking a service by selecting date,time and address. After placing the booking, the address is automatically converting to latitude and longitude by Geocoding API. Users can refresh booking history by shaking the phone.

- **Payments**

Payments are processing by using payhere with LKR.

- **Real Time Tracking**

Customer allows to track the assigned technician location through the Google map after technician assignment. Also technician can view customer location.

- **Rating and Reviews**

Customer can rate and leave a review after complete the service. It will be displaying on the detailed service fragment.

- **Notifications**

Customer and technician allows to get notification after sign in and sign up from firebase cloud messaging. Customer receives a email after technician assignment by admin.

- **Assigned bookings**

Technician can view assigned bookings on technician dashboard. Technician able to change the status of booking such as arrived and completed.

Admin

- **Authentication**

Only administrators allows to login to the admin panel.

- **User Management**

Admin allows to view user details, can ban unban.

- **Booking Management**

Administrator can assign technician to each booking, allows to change the status of the booking and can delete booking.

3.3. Functional Requirement

The functional requirements represent main operations of ServiGo system.

- The ServiGo system allows users to sign in and sign up using Firebase Authentication.
- The ServiGo system allows customers to search services.
- The ServiGo system allows customers to place a booking by choosing date, time and address.
- System shall allow to convert user address to geocode using Geocoding API.
- System shall allow customers to make payment securely with PayHere.
- ServiGo system shall store booking details only after the customer confirmation payment is successful.
- System allows users to update their profile details.
- The ServiGo shall allow technicians to view assigned bookings.
- System allows technician to change booking status (arrived, completed).
- System shall allow to track real time location of technician on google map.
- System shall allow customers to leave feedback about the service after service is completed by technician.
- System shall allow admin to login to the admin panel.
- System shall allow admin to manage user status as ban and unban.
- System shall allow admin to assign technician to each booking.
- System shall allow admin to change booking status.
- System shall allow admin to delete booking.

3.4. Non functional Requirements

- Performance : ServiGo system should respond to users actions within limited time period.
- Scalability: ServiGo should able to manage large amount of booking and users.
- Security: User personal details and transactions must handle securely and efficiently by using secure APIs and authentication.
- Reliability: ServiGo should run and operate continuously without any issue.
- Usability: ServiGo application should provide user friendly, modern and simple interface.
- Availability: The ServiGo application should accessible in anytime.
- Maintainability: ServiGo system should be easy to update and maintain.

- Compatibility: ServiGo application shall be run in devices that API level is 28 and above.

4. Design

4.1. System Architecture

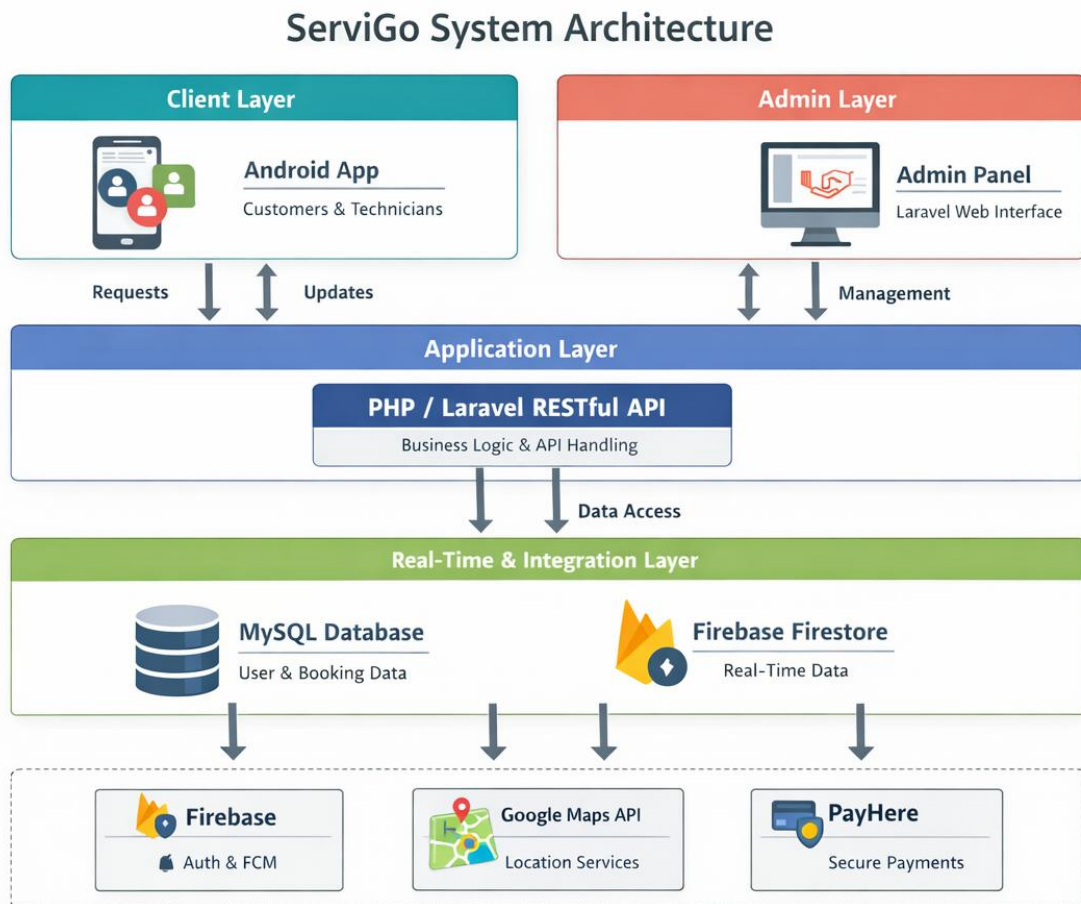


Figure 1: System Architecture

1. Client Layer

This includes mobile application used by customer and technician. In this layer customer can brows service, book service, make payments and track locations. Technician can view customer location, change the status of booking. This layer sends requests to the backend for the specific operations.

2. Admin Layer

This layer represent by laravel application (Admin panel) that use MVC architecture. Admin can ban and unban users, assign technicians to bookings, update booking statuses, delete bookings and oversees the system from dashboard.

3. Application Layer (Backend API)

The backend of ServiGo is developed by using PHP and RESTfull architecture. It is handling bussines logics, handle client requests, make communication between mobile application, mysql database, firebase firestore and admin panel.

4. Data and Real Time Layer

This layer contained all data of the system. MySQL database stored such as admins, bookings, reviews. Firebase firestore stored users, technician locations, services. Firebase storage is used to store banner pictures and profile picture of the users.

5. Integration Layer

This layer include third party services such as firebase, google map APIs, PayHere.

4.2. Technology Stack

Layer	Technology
Mobile App	Java (Android), XML Layouts, Material Design3, ViewBinding, Advanced Components, Telephony, Sensors (Motion, GPS)
Backend	PHP REST API, MySQL
Admin Panel	Laravel Framework(11, PHP 8.5v), Blade Templates, Eloquent ORM, Tailwind CSS
Real Time Services	Firebase Authentication, Firestore, Cloud Messaging, Storage
Maps & Location	Google Maps SDK, Geocoding API, Play Services Location
Payments	PayHere SDK
Media Handling	Glide, Picasso
Networking	Retrofit 2 (Gson and Scalars)

Table 1: Technology Stack

4.3. Project Architecture

Package	Description
---------	-------------

lk.teveram.servigo.activity	Main Activity entry points (Splash, Login, Maps, Dashboards).
lk.teveram.servigo.fragment	UI components (Home, Booking Form, Profile, Service Details).
lk.teveram.servigo.api	API client configurations and endpoint definitions.
lk.teveram.servigo.model	POJO classes for data mapping (User, Booking, Review, ServiceCategory).
lk.teveram.servigo.adapter	RecyclerView adapters (Bookings, Services, Reviews).

Table 2: Project Architecture

4.4. Use Case Diagram



Figure 2: Use Case Diagram

4.5. Class Diagram

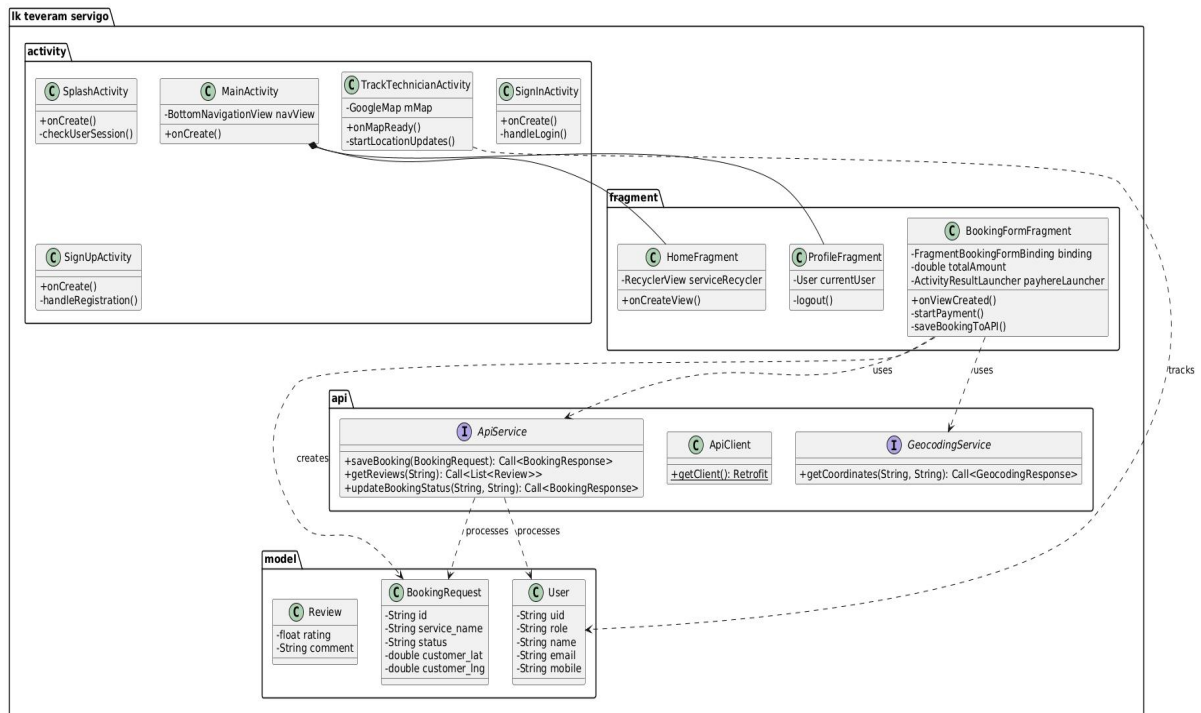


Figure 3: Class Diagram

4.6. Sequence Diagram

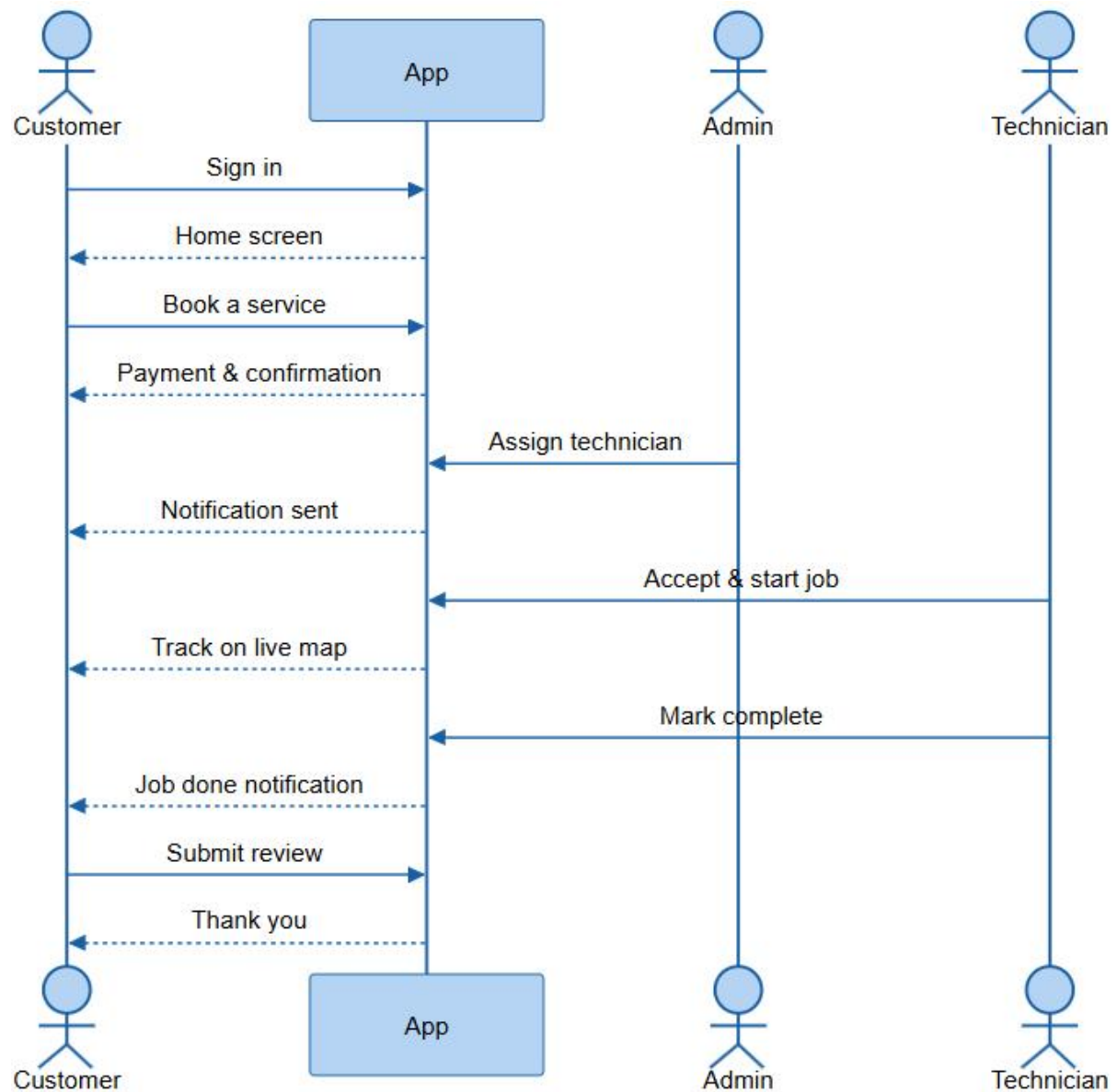


Figure 4: Sequence Diagram

4.7. ER Diagram (MySQL)

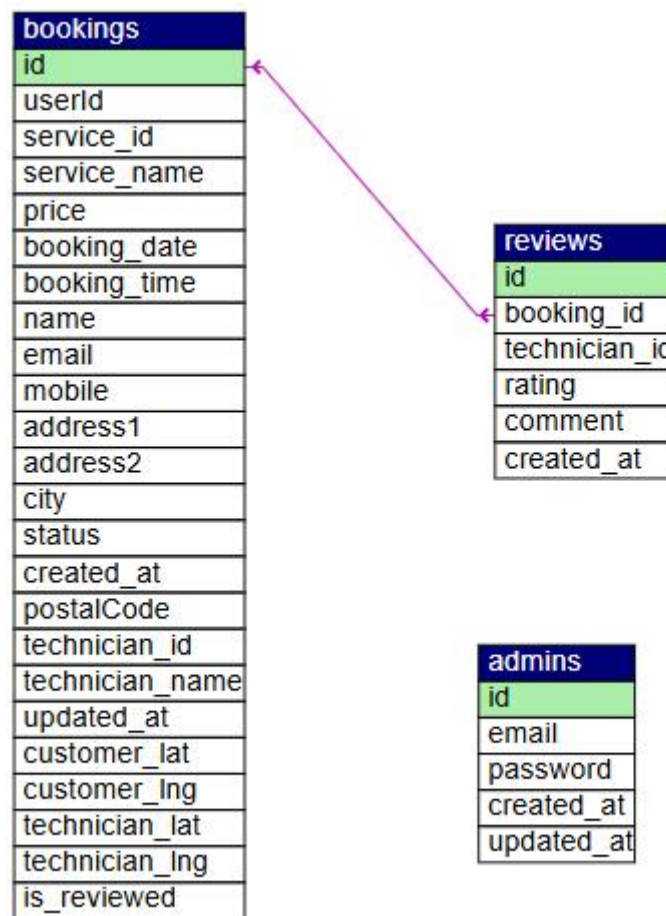


Figure 5: MySQL ER Diagram

4.8. ER Diagram (Firebase Firestore)

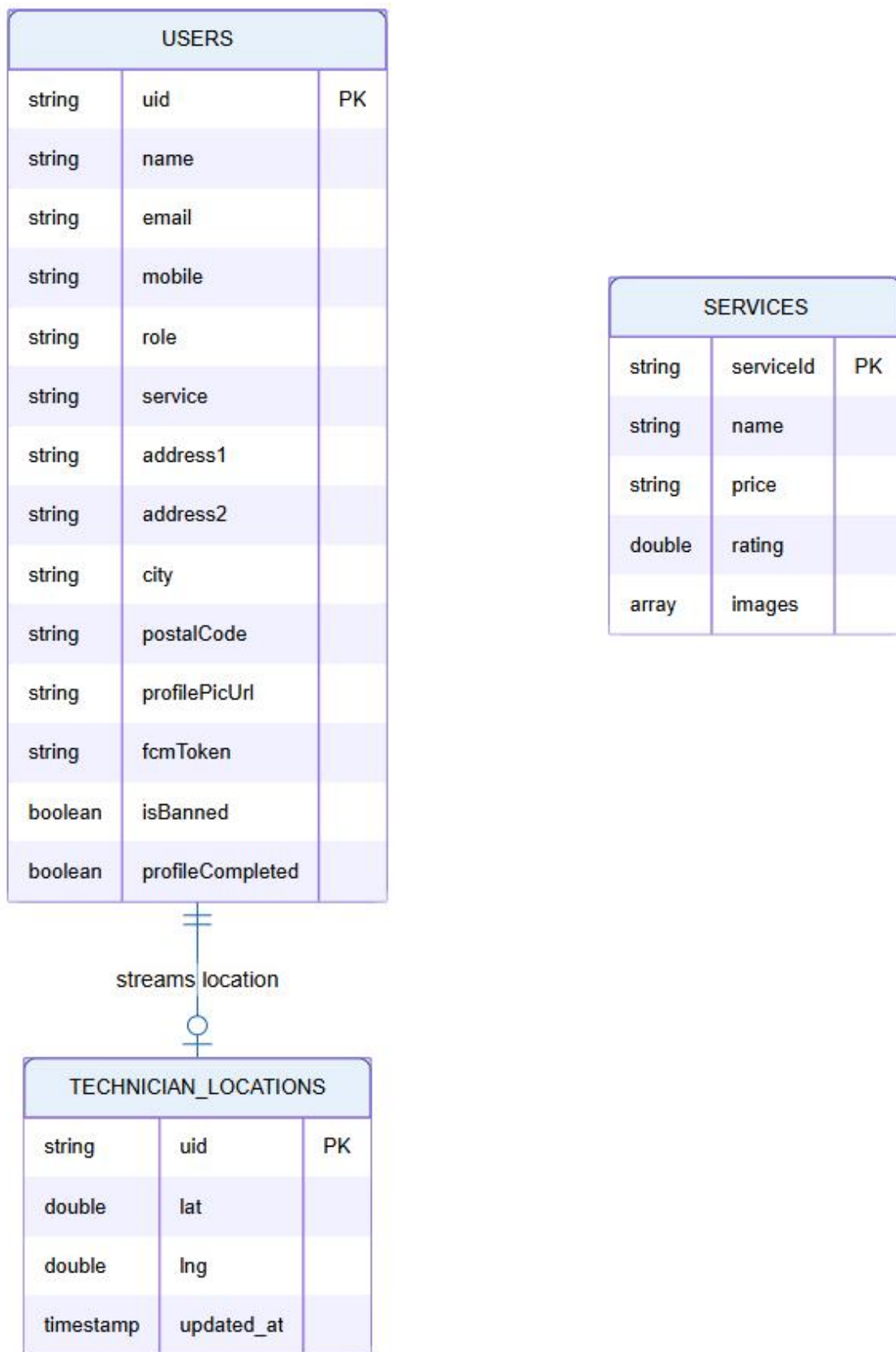


Figure 6:ER Diagram (Firebase)

4.9. Combined ER Diagram (MySQL + Firebase Firestore)

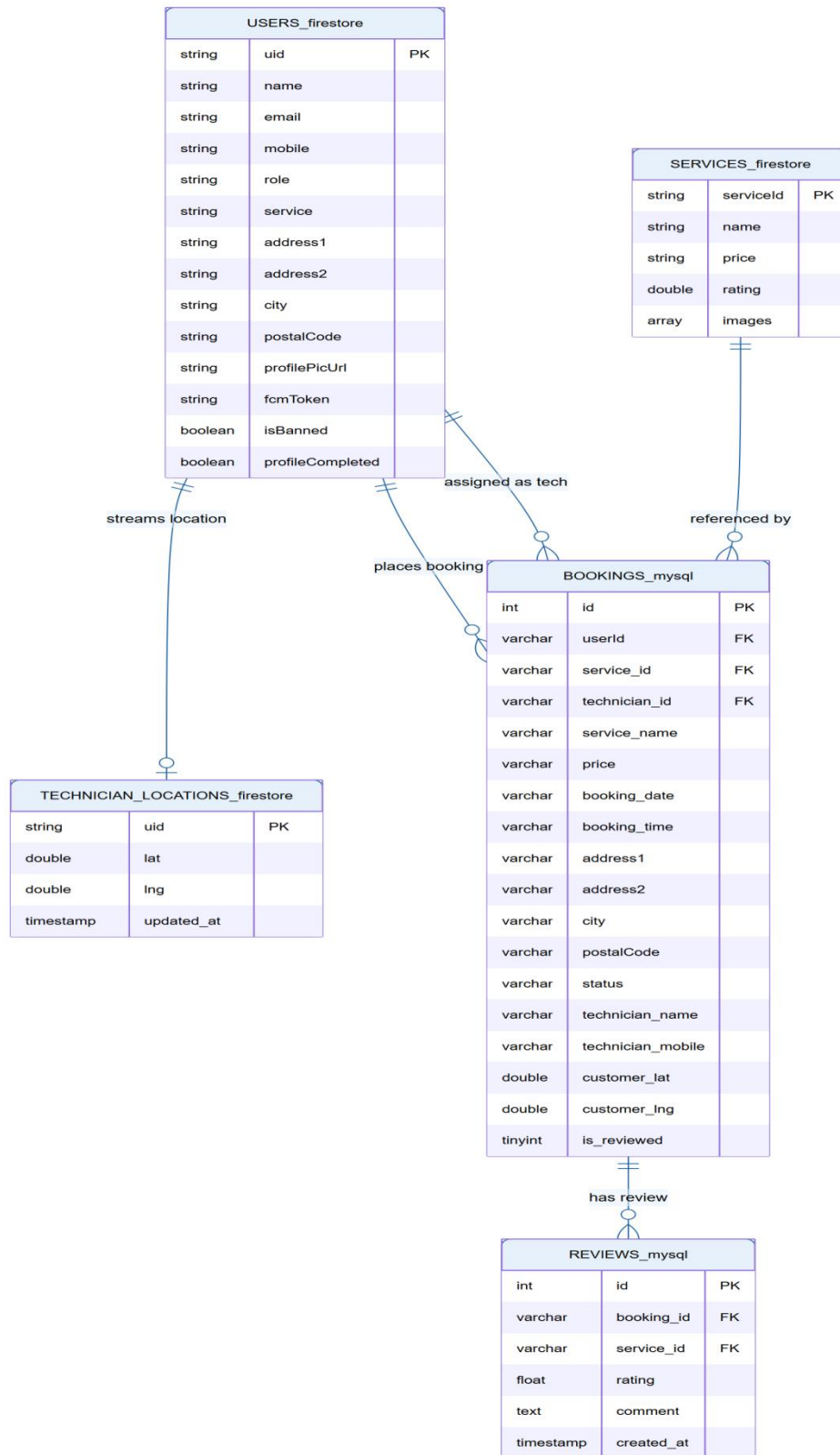


Figure 7: Combined ER Diagram (MySQL + Firebase)

4.10. User Interface (UI) Design

Splash Screen

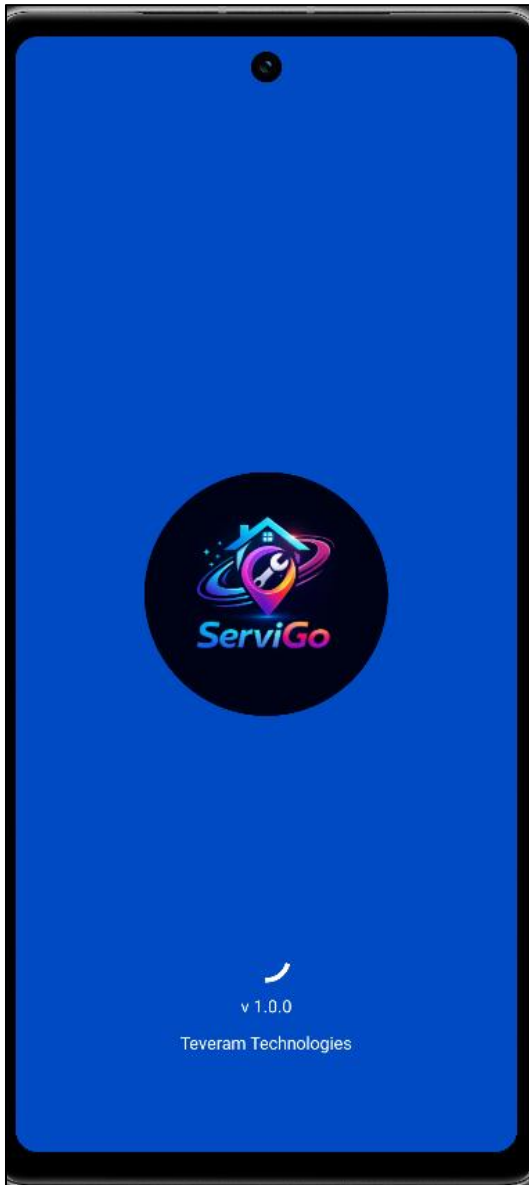


Figure 8 : Splash Scree Light Theme

This is the Splash Screen of the Servigo. The Application starts with this screen.



Figure 9 : Splash Screen Dark Theme

Sign In Screen

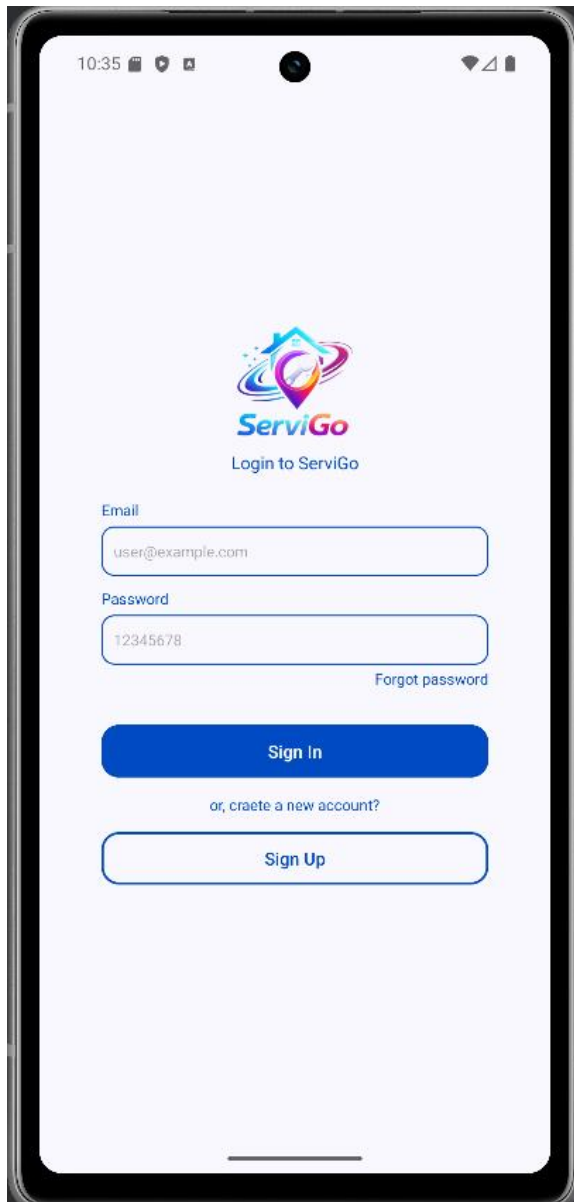


Figure 10: Sign in Light theme

This is the Sign screen of the ServiGo. After the splash screen user can see this screen and if user already registered, user can sign in from here. If user is new to ServiGo user can touch Sign Up button and it redirects to sign up screen.

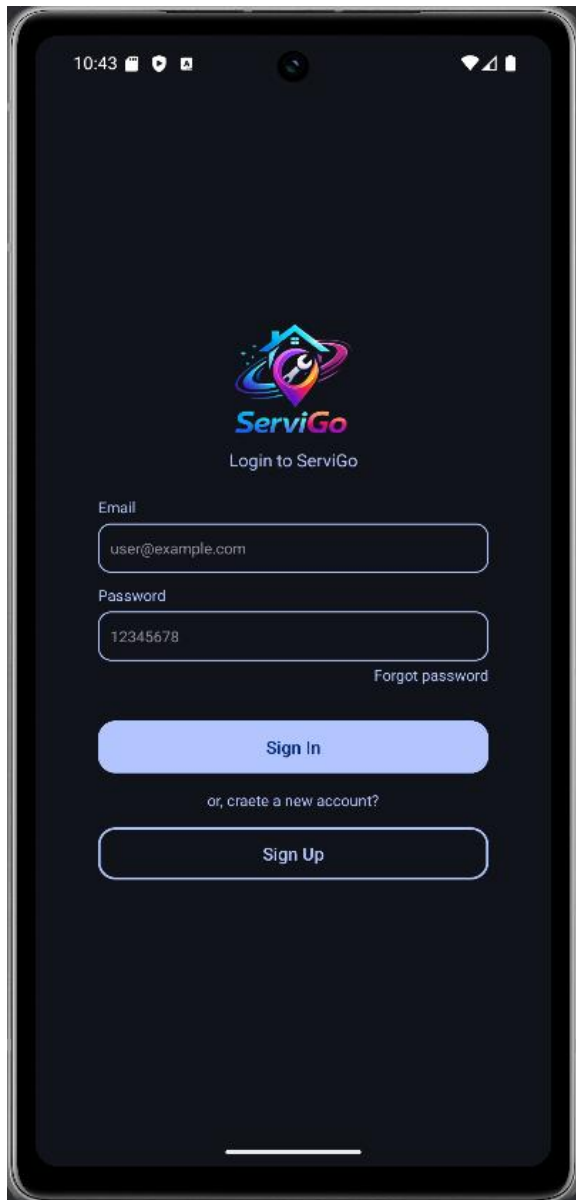


Figure 11: Sign In Dark Theme

Sign Up Screen

10:43

ServiGo

Create Account

Name

Randy Joe

Email

user@example.com

Password

Password must contain at least 8 characters

Retype Password

Customer Technician

Sign Up

Already have an account ?

Sign In

Figure 12 : Sign Up Screen Light

This is the Sign Up Screen of the ServiGo App. After clicking Sign up button on Sign In screen, user will see this screen. In this screen user can select the role and must enter the details to create new account. All the input fields are validated.

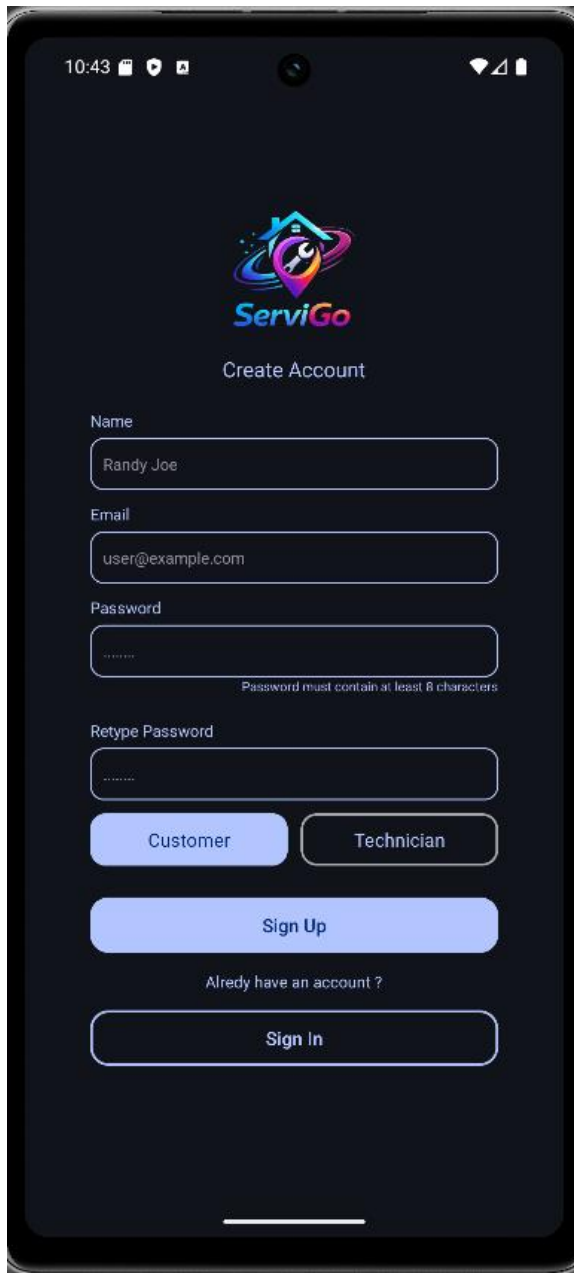


Figure 13 : Sign Up Screen Dark

Customer Home Screen

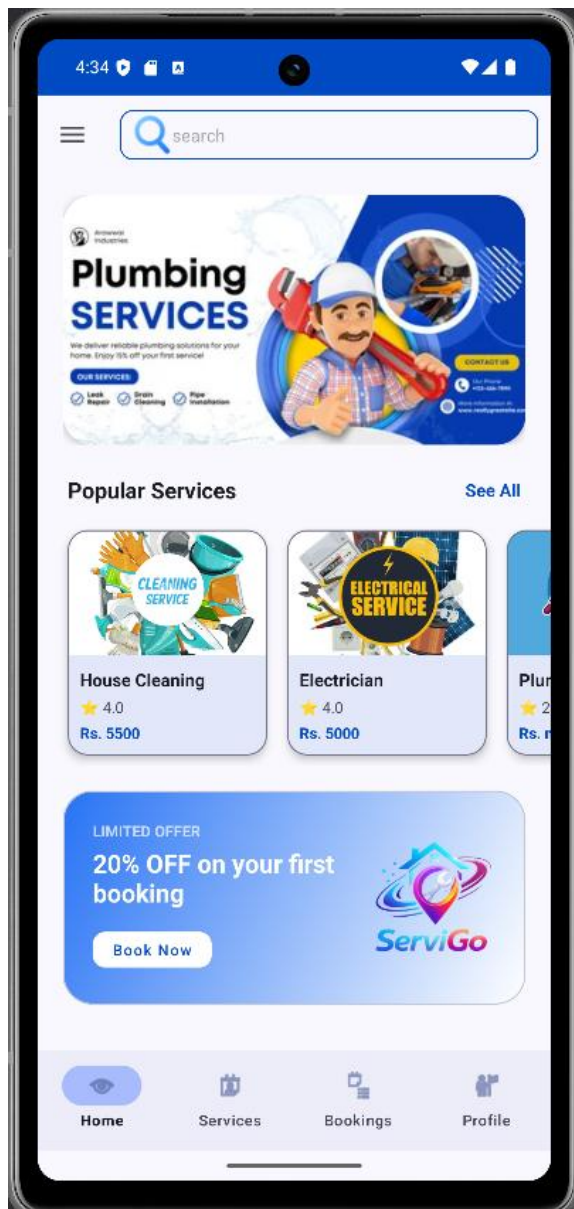


Figure 14: Customer Home screen Light theme

After customer sign in, customer can view the home screen of ServiGo application. In this screen, customer can see automated image slider, popular services based on customer ratings, banner. With scrolling down customer can see all services section. In the header there is a search bar the allows user to search any services. After search services it will show in this screen. The hamburger icon of the menu leads to open side navigation bar. Customer allows to navigate between screens by using bottom navigation bar.

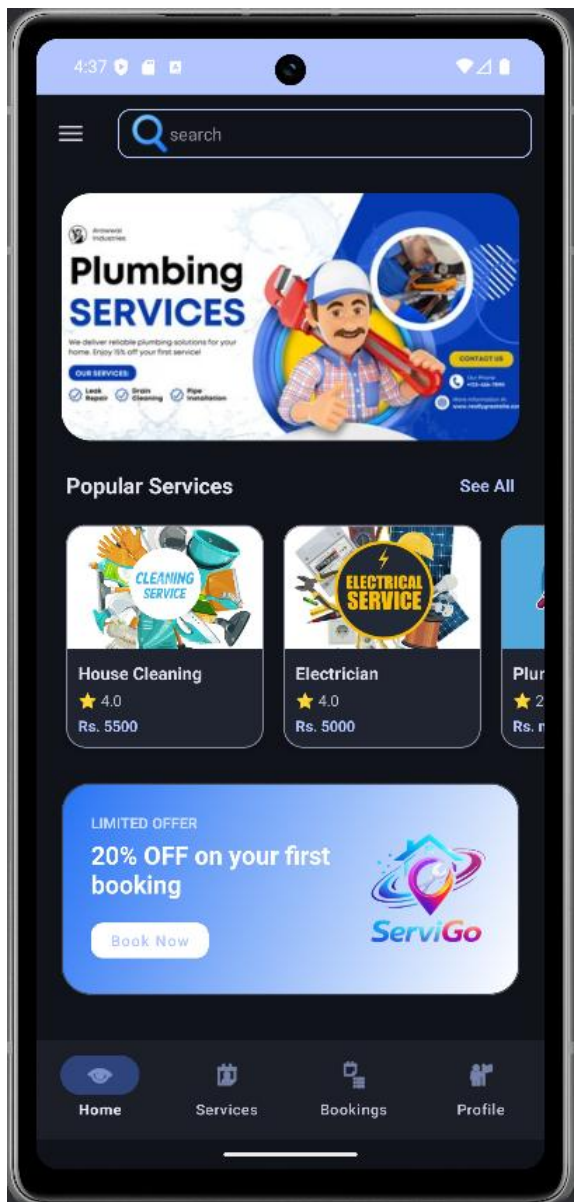


Figure 15: Customer Home screen Dark theme

Side Navigation Bar

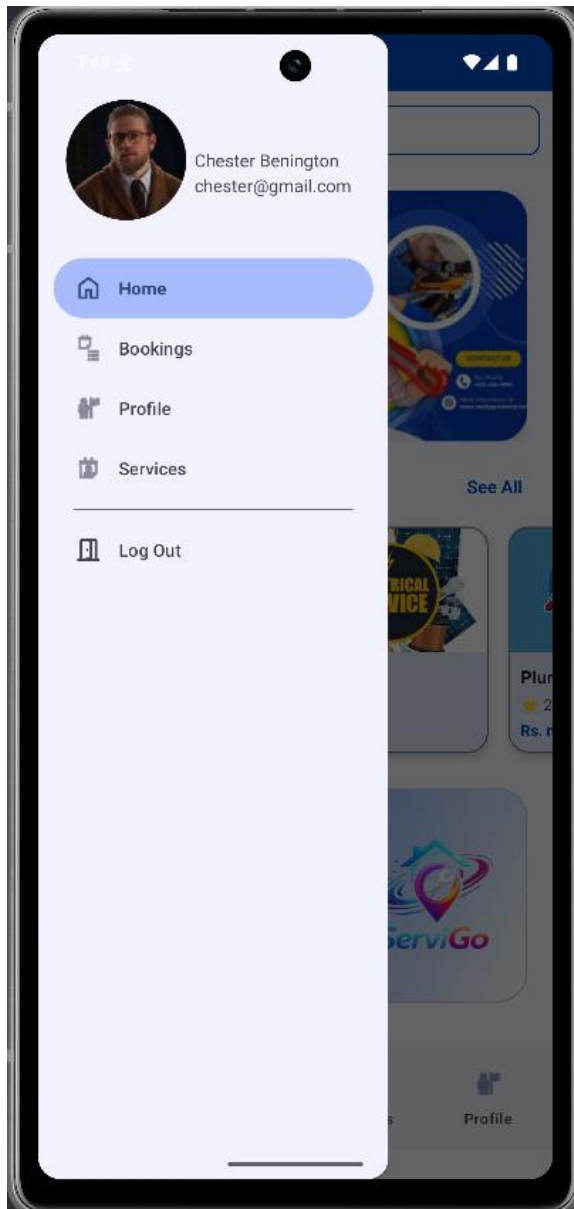


Figure 16: Side navigation bar Light Theme

This is the side nav bar of ServiGo. It shows navigation buttons to Bookings, profile, Services. Also user can logout by logout button. Top of the side bar shows user full name, email and the profile picture.

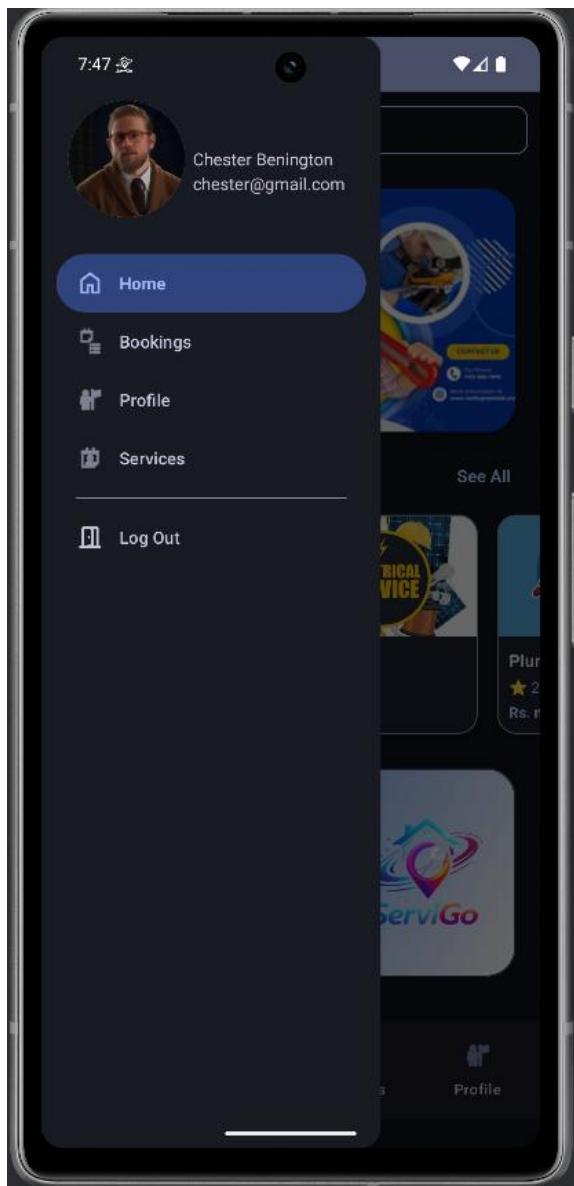


Figure 17: Side navigation bar DarkTheme

Services Screen

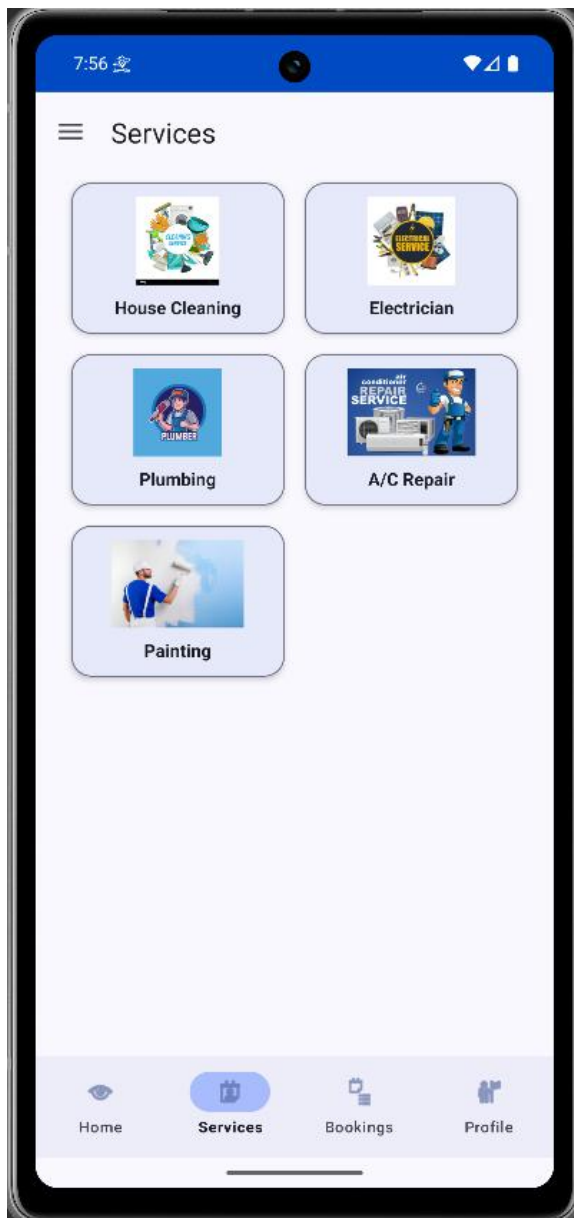


Figure 18: Services Screen Light Theme

In this screen customer able to see all the services and allows to view details of each services by touching s service.

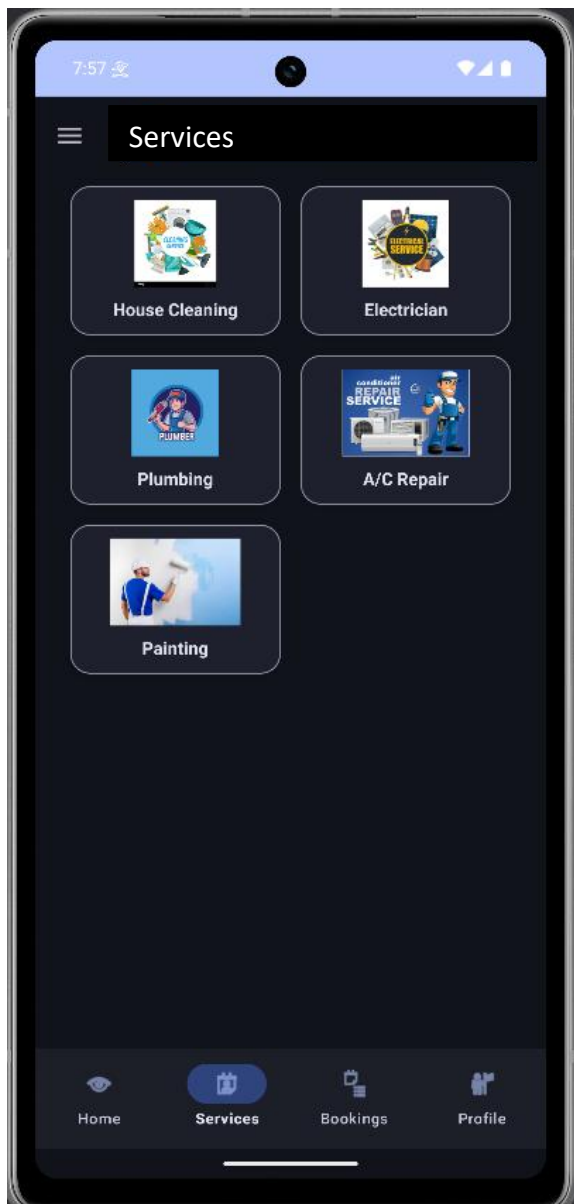


Figure 19: Services Screen Dark Theme

Single Service Screen

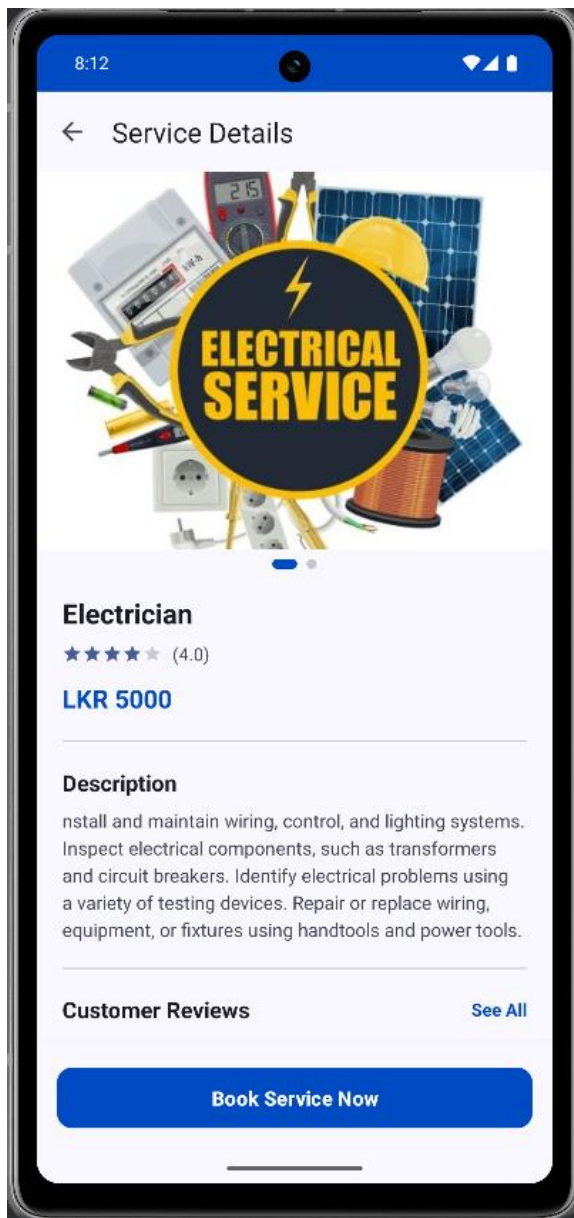


Figure 20: Single service screen light theme

This screen shows details of the service, reviews of customers and ratings. User can fill the booking form by touching book service now button.



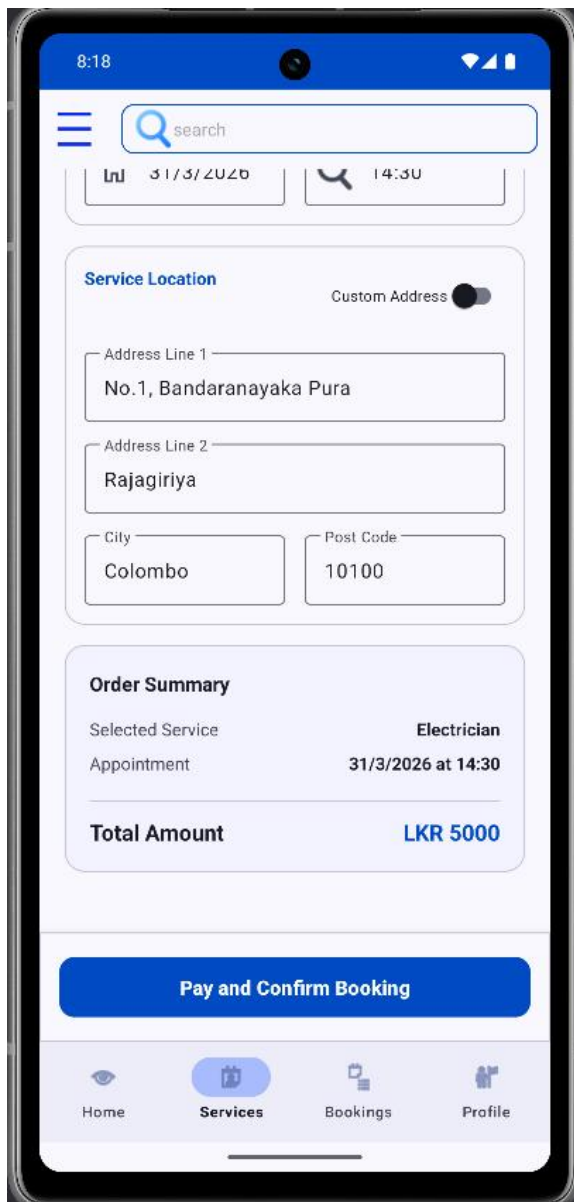
Figure 21:Single service screen light theme

Booking Form screen

The screenshot shows a mobile application interface for a service booking form. At the top, there is a blue header bar with the time 8:18 and status icons. Below the header is a search bar with a magnifying glass icon and the word "search". The main title "Service Booking" is displayed in bold. The form is divided into three sections: "Contact Information", "Appointment Schedule", and "Service Location". The "Contact Information" section contains three input fields: "Full Name" (filled with "Chester Benington"), "Email Address" (filled with "chester@gmail.com"), and "Contact Number" (filled with "0776666963"). The "Appointment Schedule" section contains two input fields: "Date" (filled with "31/3/2026" and a calendar icon) and "Time" (filled with "14:30" and a magnifying glass icon). The "Service Location" section contains a "Custom Address" toggle switch (turned on) and an "Address Line 1" input field. A large blue button labeled "Pay and Confirm Booking" is positioned below the form. At the bottom, there is a navigation bar with four icons: "Home", "Services" (highlighted), "Bookings", and "Profile".

Figure 22:Booking From

This is the booking form. The bookings are storing based on this details. Customer must fill this form to confirm the booking. Customer can add new address to the booking by toggling the custom address toggle.



Payment Phase

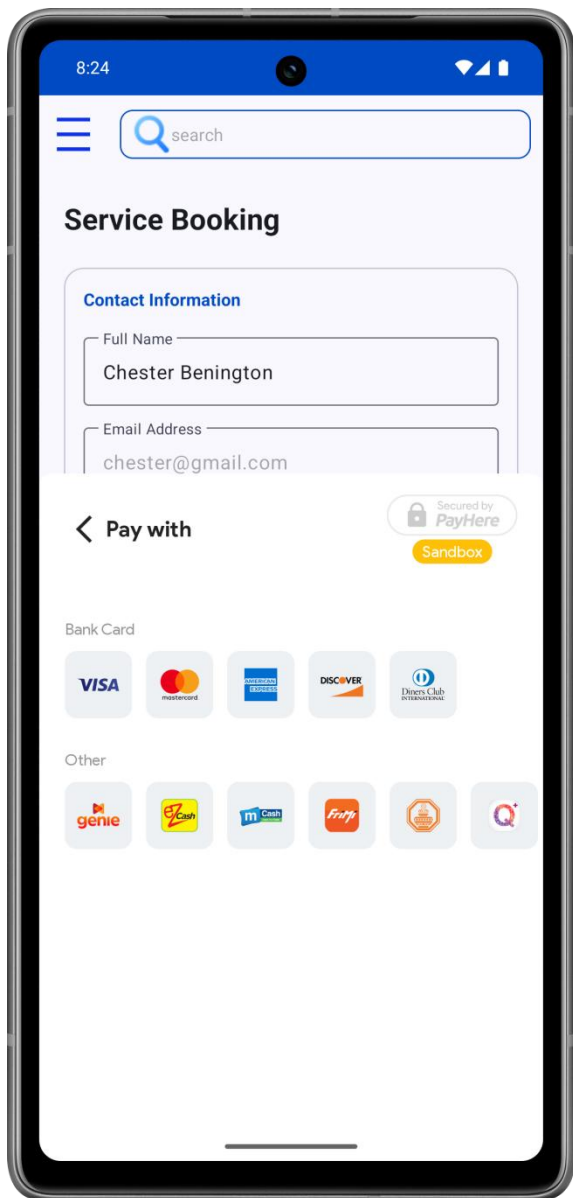
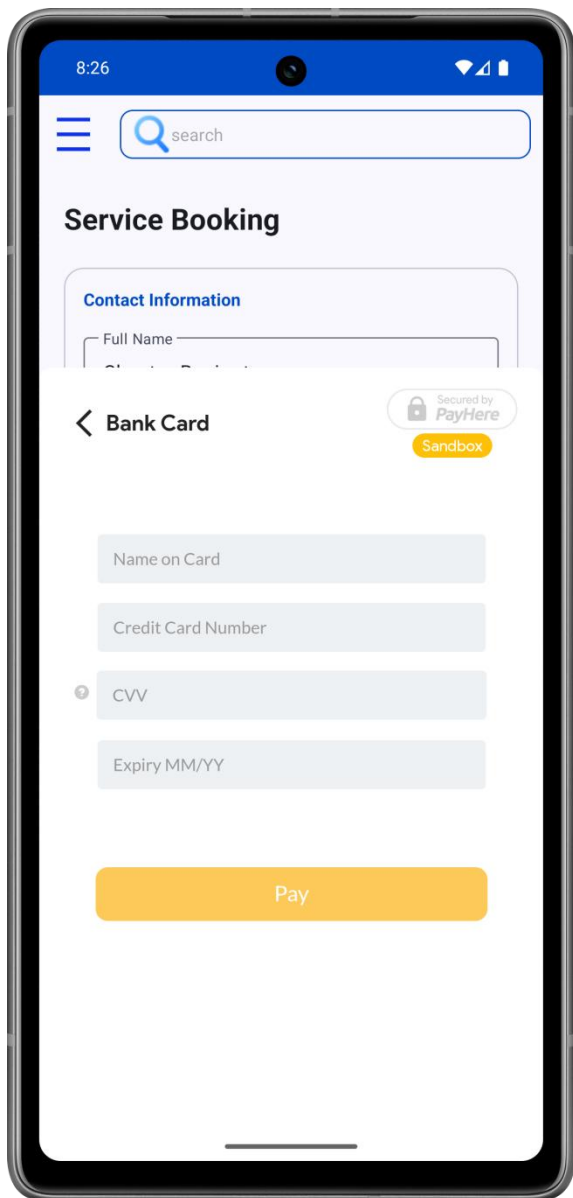


Figure 23: Payment Phase

User allows to fill the card details and make the payment.



Payment confirmation and storing booking

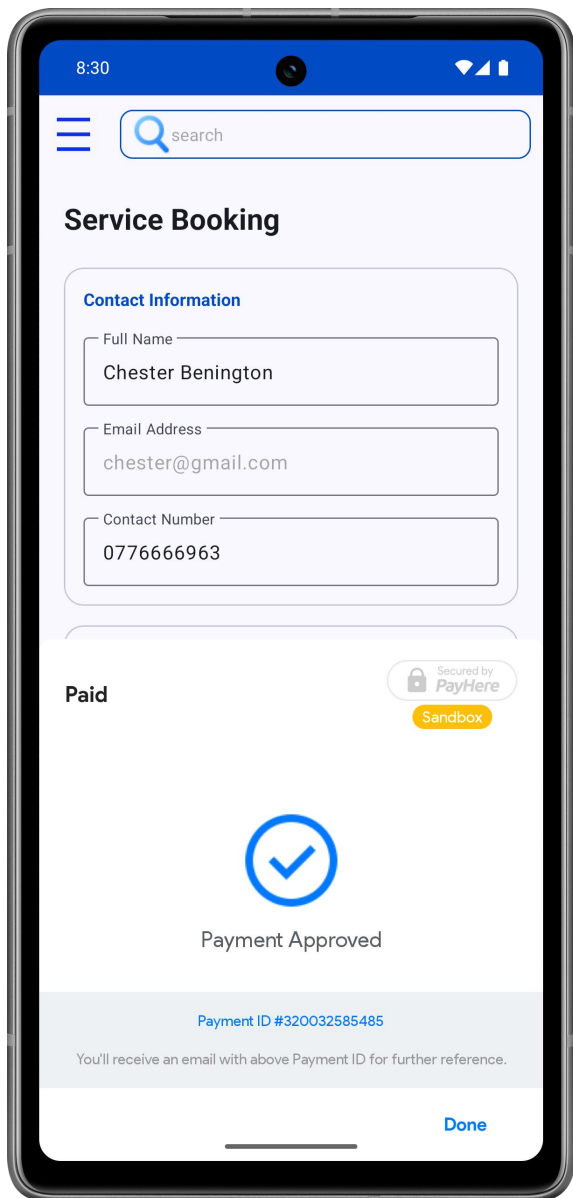


Figure 24 Payment confirmation

Profile Screen

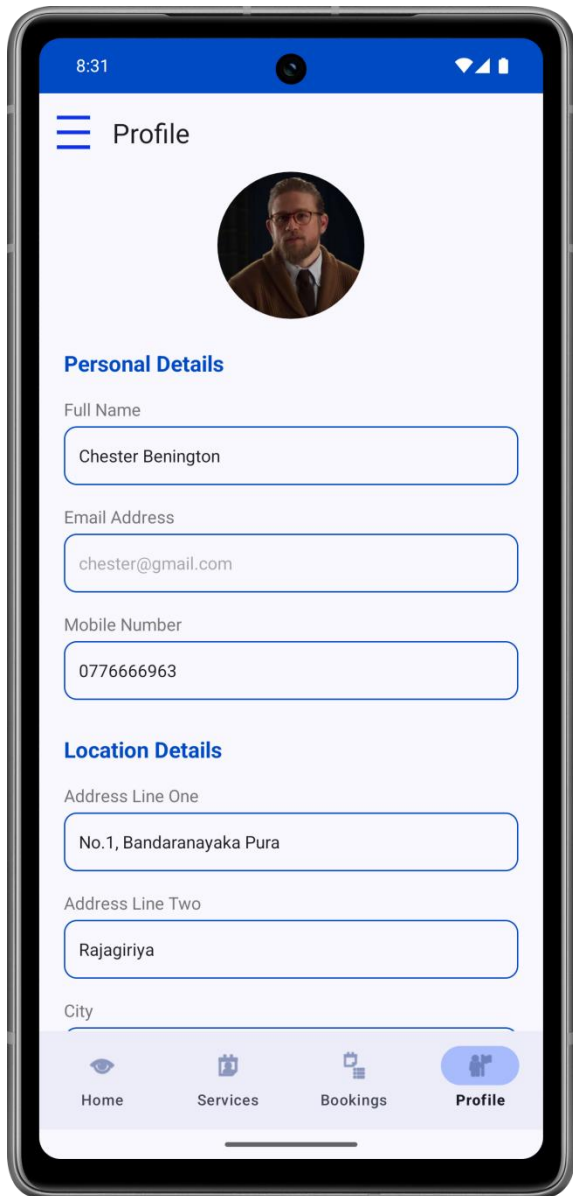
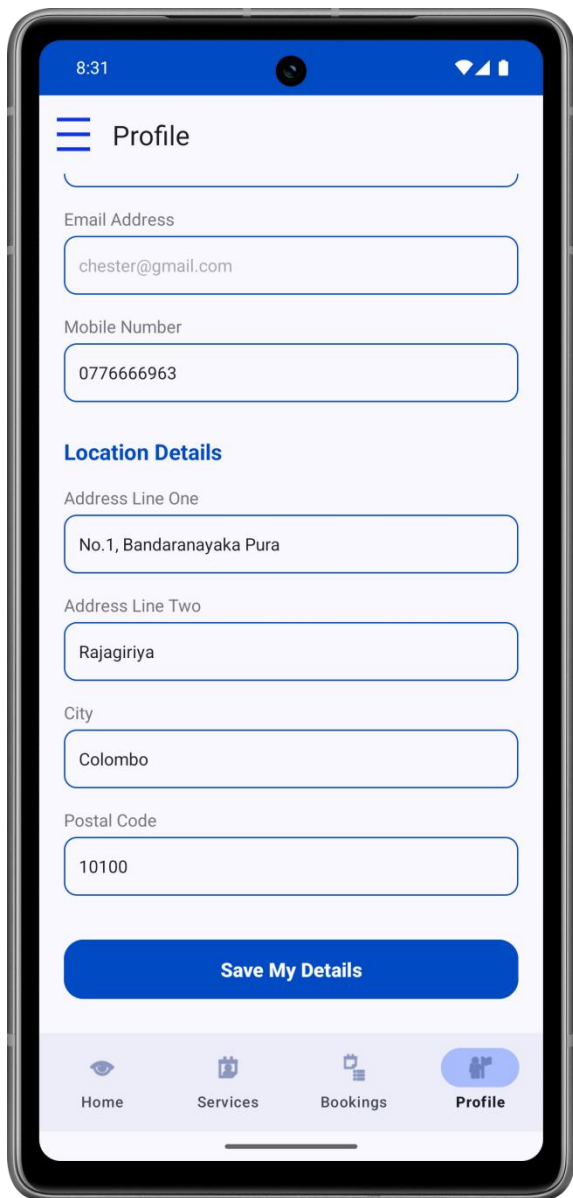


Figure 25: Profile Screen

In this screen user can view their personal details and user allows to update their details.



Booking History

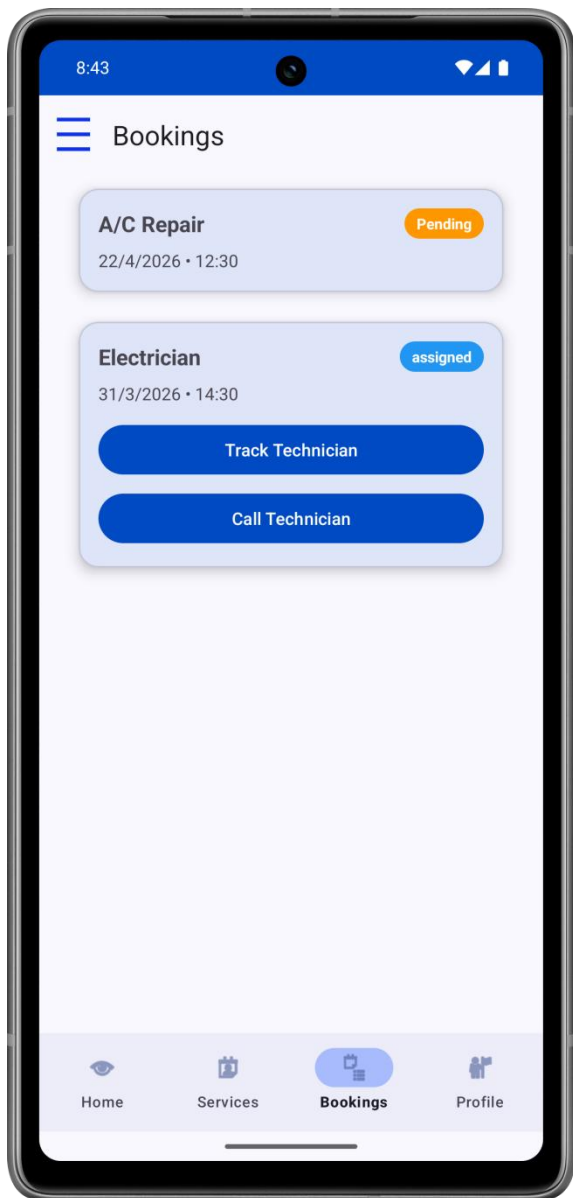


Figure 26: Booking History

Customer can see booking history and status. After technician assignment, customer allows to track the technician and call to technician. When customer shake the mobile, booking list will be refresh.

Map

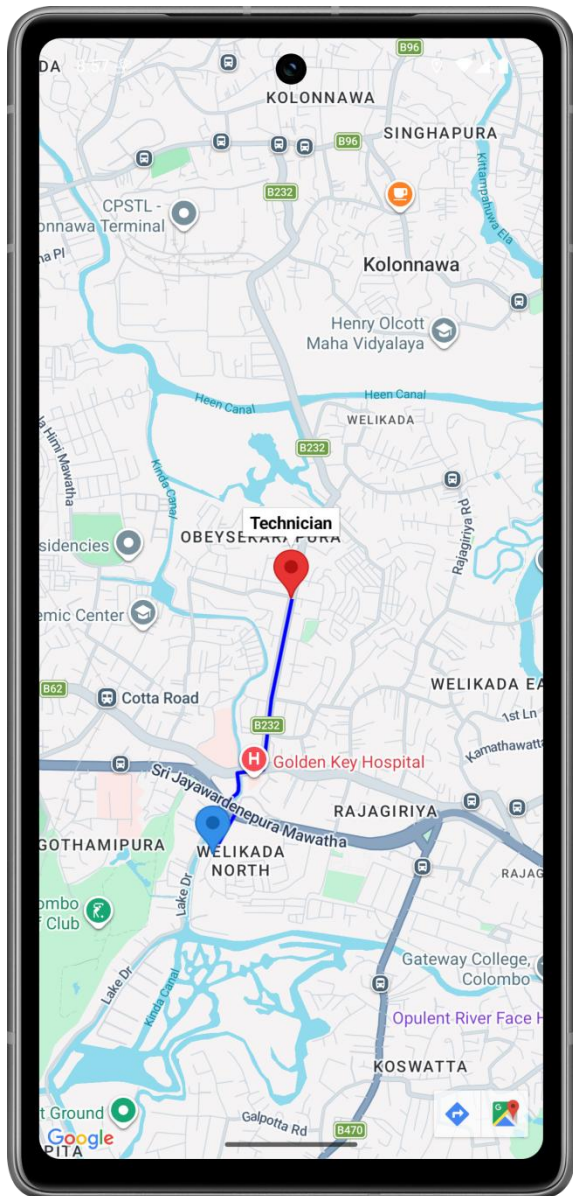


Figure 27: Map

Customer can track technician on Google map. The marker of technician is moving in real time when technician moves.

Review Section

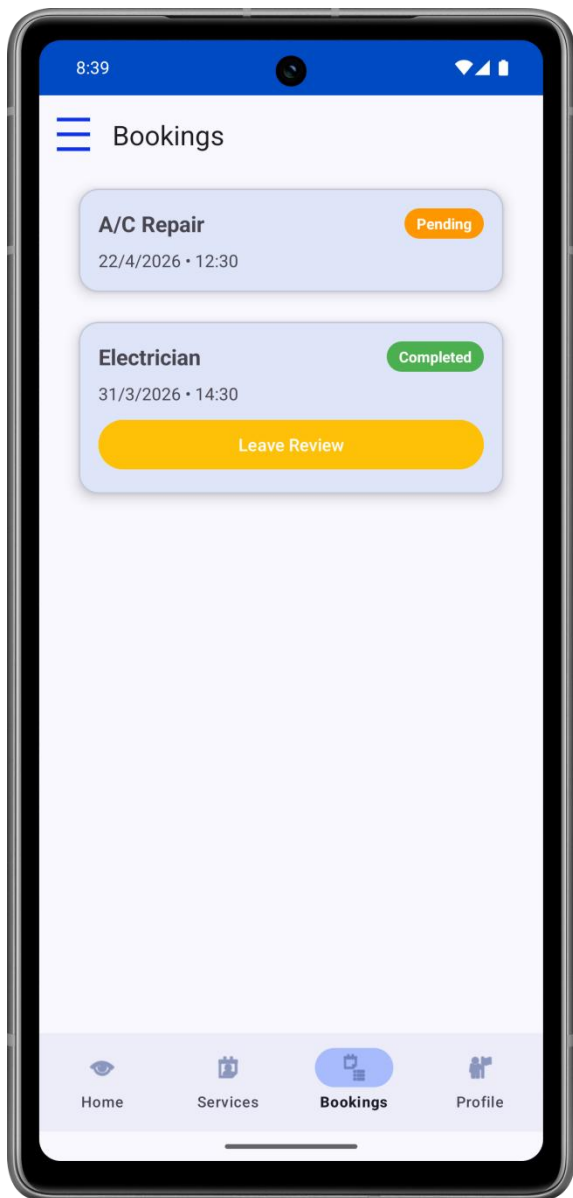
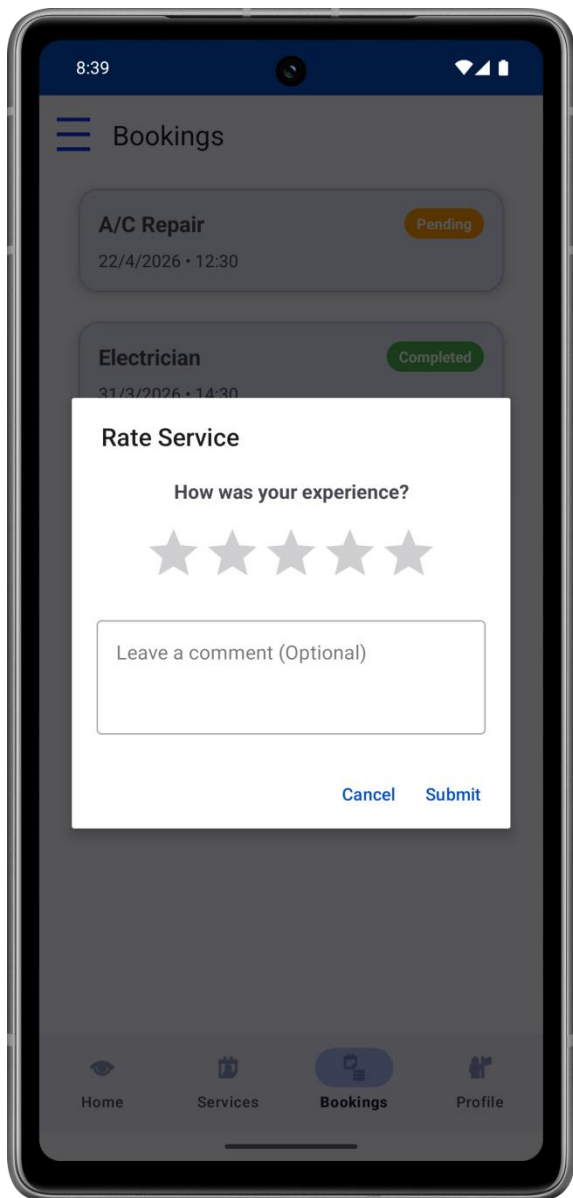
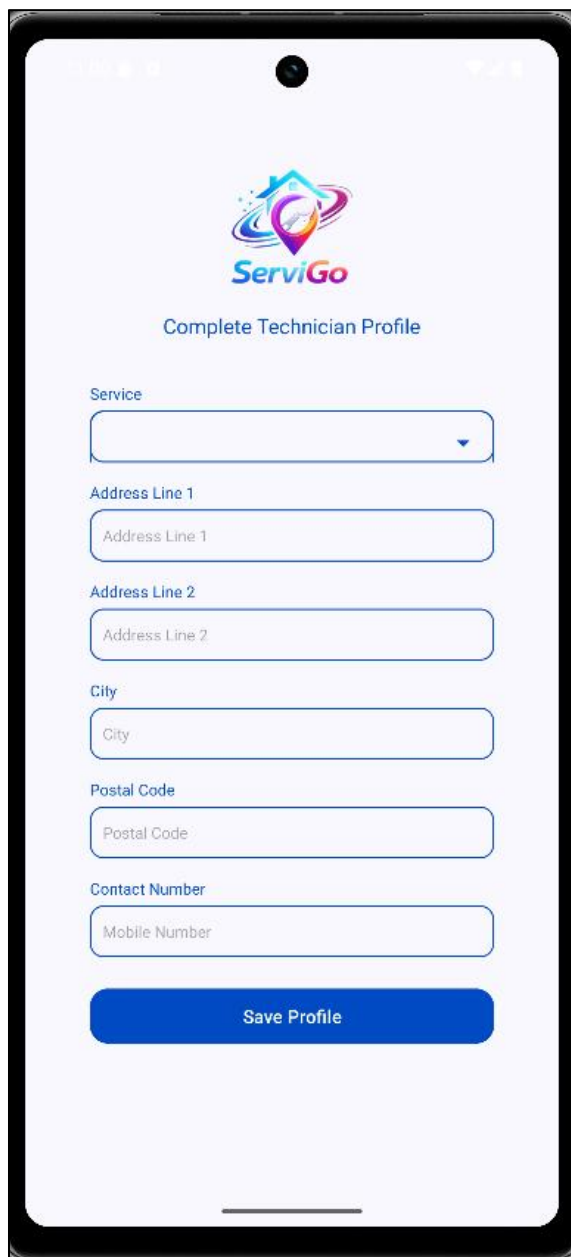


Figure 28: Review Section

When the service is completed, customer can rate the service and leave a comment.



Complete Technician Profile Screen



The screenshot displays the 'Complete Technician Profile' screen of the ServiGo app. At the top, the ServiGo logo is centered above the title 'Complete Technician Profile'. Below the title, there are several input fields: a dropdown menu for 'Service', two text boxes for 'Address Line 1' and 'Address Line 2', a text box for 'City', a text box for 'Postal Code', and a text box for 'Contact Number' (labeled 'Mobile Number'). A blue 'Save Profile' button is positioned at the bottom of the form. The screen is framed by a black border, and the status bar at the top shows the time as 10:10 and battery level at 44%.

Figure 29 : Complete Technician Profile Screen Light Theme

This is the Complete Technician Profile Screen of the ServiGo App. After the sign up as technician, user must create profile with details such as service, address, city, postal code and contact number.

11:01

ServiGo

Complete Technician Profile

Service

Address Line 1

Address Line 2

City

Postal Code

Contact Number

Mobile Number

Save Profile

Figure 30 : Complete Technician Profile Screen Dark Theme.

Technician Home

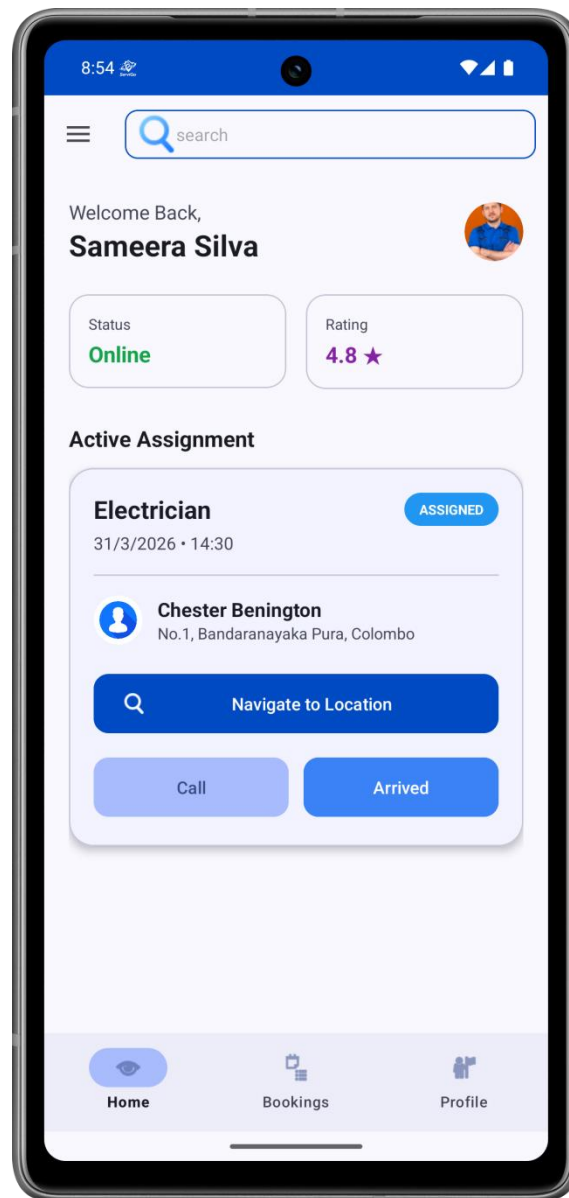


Figure 31: Technician Home

Technician can view assigned bookings from home screen, and can view the customer location on Google map. Technician allows call customer. After technician arrived, technician should press arrived button. Then booking status will be change.

Booking History

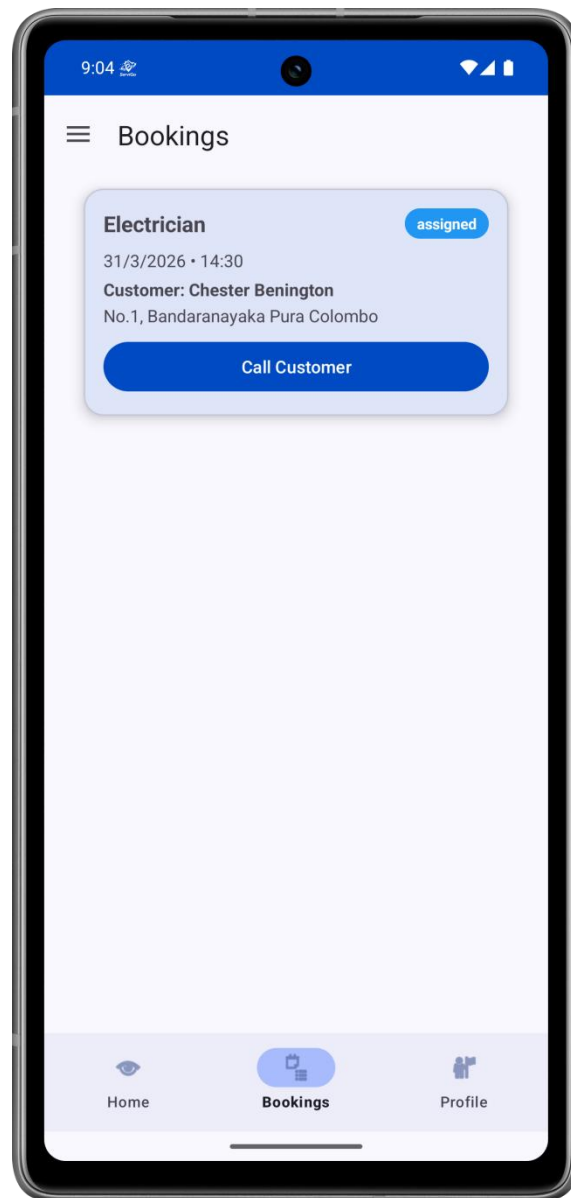


Figure 32: Technician booking history

When technician shake the mobile, booking list will be refresh.

Status Changing of booking

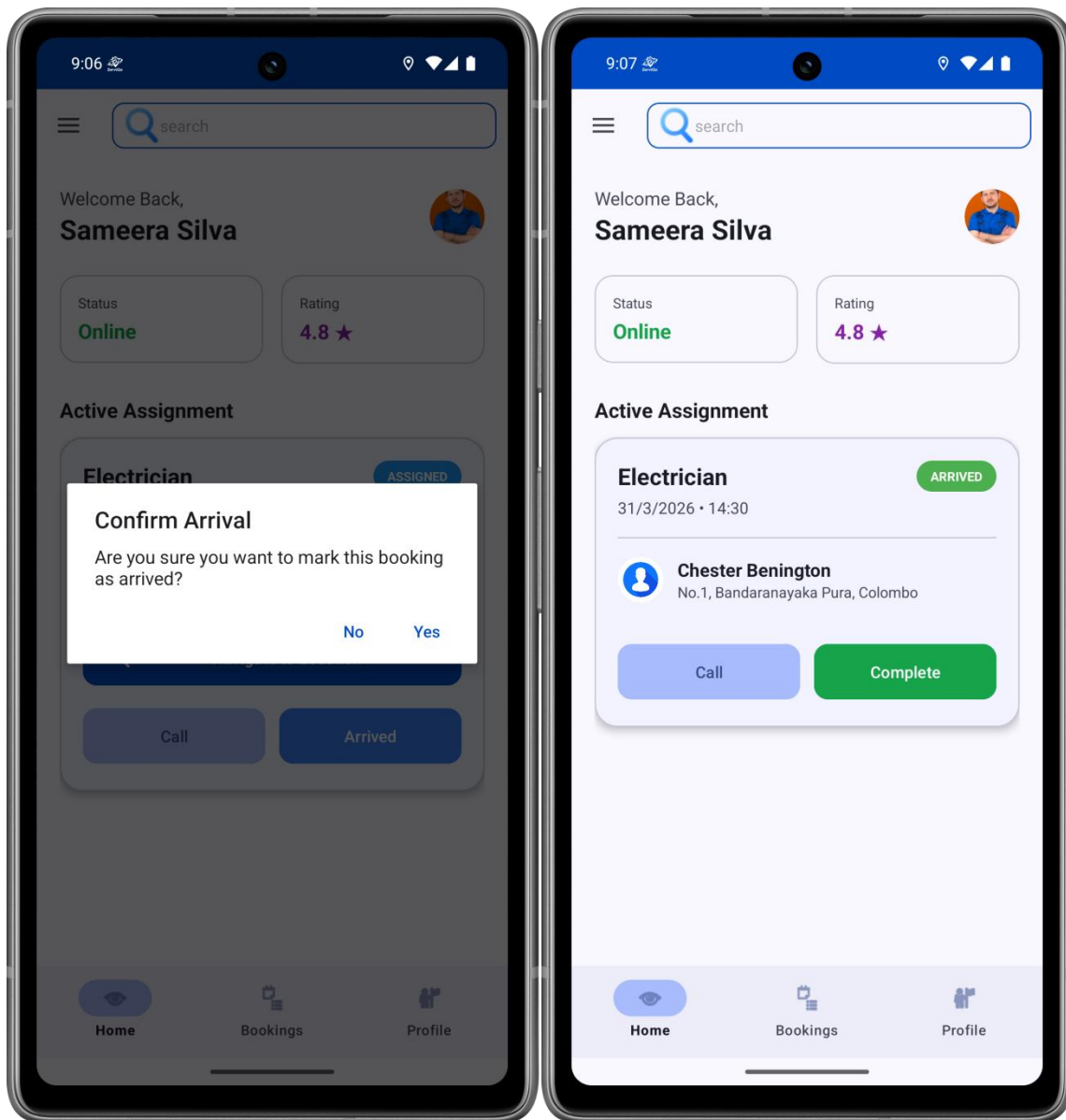


Figure 33: Status Changing

After technician arrived, technician should press arrived button. Then booking status will be change and show the complete button to press.

Admin Login

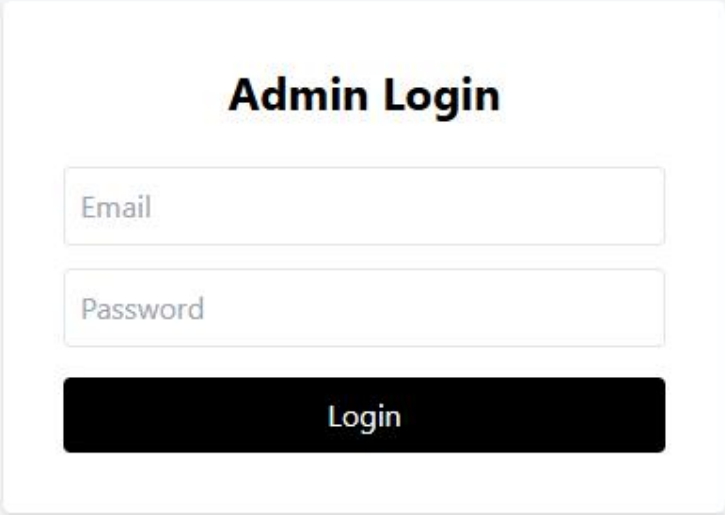
The image shows a web form for Admin Login. It is centered on a light gray background. The form itself is a white rectangle with a thin gray border. At the top of the form, the text "Admin Login" is written in a bold, black, sans-serif font. Below this title, there are two input fields. The first field is labeled "Email" in a light gray font and has a white background with a thin gray border. The second field is labeled "Password" in a light gray font and also has a white background with a thin gray border. Below these two fields is a solid black rectangular button with the word "Login" written in white, centered text.

Figure 34: Admin Login

Admin can log in to the laravel based admin panel using this page.

Booking Management

ServiGo

Dashboard

Users

Logout

Dashboard

Total

7

Pending

4

Completed

1

Search...

mm/dd/yyyy

ID	Service	Customer	Date	Technician	Status	Actions
2	A/C Repair	Devin Liyanapathirana 0943666999	21/3/2026	—	Pending	Assign Status Delete
7	Painting	Tom Jay 0776999636	26/3/2026	—	Pending	Assign Status Delete
8	Painting	Tom Jay 0776999636	31/3/2026	—	Pending	Assign Status Delete
9	Painting	Tom Jay 0776999636	31/3/2026	Nimal Silva	Completed	Assign Status Delete
10	Painting	Tom Jay 0776999636	28/3/2026	Nimal Silva	assigned	Assign Status Delete
14	Electrician	Chester Benington 0776666963	31/3/2026	Sameera Silva	arrived	Assign Status Delete
15	A/C Repair	Chester Benington 0776666963	22/4/2026	—	Pending	Assign Status Delete

Figure 35: Booking management(Dashboard)

Admin able to manage bookings such as, update status, technician assignment and delete bookings. In this page, admin can view count of all bookings, completed bookings and pending bookings. After technician assignment, customer get email of assignment.

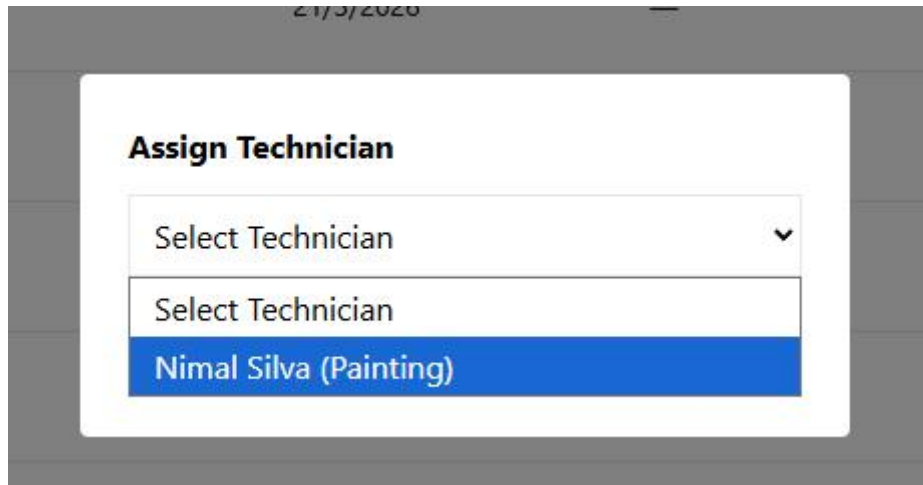


Figure 36: Technician Assignment

This pop up loads filtered technicians based on booking service

User Management

ServiGo

Dashboard

Users

Logout

User Management

Refresh List

Total Users
6

Customers
3

Technicians
3

ID	Name	Email & Mobile	Role	Status	Actions
CucIm601vChbZASpmHXINBoIPvF3	Nimal Silva	nimal@gmail.com 0778558585	Technician	Banned	Unban
Su77WoCa1gA7K3uO2UhfqNkTCo2	Chester Benington	chester@gmail.com 0778666993	Customer	Active	Ban
VhHcbrTA9eYsoAR45kz19Uj6fk1	Tom Jay	tom@gmail.com 0778999636	Customer	Active	Ban
dbH69TmM8hSbdIDwU3Jz5ZVV2	Prasanna Jayawardena	prasanna@gmail.com	Technician	Active	Ban
g9UC7R9C3XPOs6bLz8nP2zOMBqn1	Devin Liyanapathirana	devin@gmail.com	Customer	Banned	Unban
uYcJ8dv8e0a7aXb5BRKqkDkmu53	Sameera Silva	sameera@gmail.com 0740345236	Technician	Active	Ban

Figure 37: User management

Admin can see total users, customer count and technician count. Admin allows to ban an unban users from this page.

5. Implementaion

5.1. Backend Implementation

To develop the backend of ServiGo uses PHP and follows the RESTful API. It helps to improve system central performance of ServiGo. This backend is handles all the requests from the client and responses, It communicates with mobile application, laravel based admin panel, MySQL database , Firebase Firestore and handles business logics.

This backend is used to handle some main operations of the ServiGo system such as store bookings, get booking details, updates status of bookings (in Technician side and admin side), technician assignments and delete bookings. Additionally the backend used to change users status like ban and unban. The ServiGo system shares data by using HTTP methods, like POST and GET. And it shares data in JSON format.

In additionally, the admin panel of ServiGo is developed with backend by using Laravel framework and it follows MVC architecture. It enable admins to assign technicians to each booking, update status of booking, delete booking, ban and unban users. In this phase represent this project is uses clean and well structured backend API and that ensures efficiency, scalability of the ServiGo system. And is is helps to communicate with other services.

5.2. Frontend Implementation

The main front end is mobile application. The mobile application is designed by using java. The interfaces of mobile application is designed by using XML layouts with material design models. This combination is ensures the ServiGo provide modern, clean and smooth frontend.

The frontend of this project uses fragment based architecture with activity containers. With this architecture, user can navigate fast between other screens and it helps to manage better memory.

The mobile application that developed by using java is the frontend of the servigo project. The interfaces of serviGo is developed by using XML layouts with material design principles. This ensures that the interfaces gives are clean and user friendly experience. Frontend used RecyclerView for service listing, ViewPager2 for image slider, Customized input fields for user friendly interfaces. This ensures this ServiGo front end display dynamic data by using that components. Also View Binding is used to interaction with UI elements.

5.3. Database Design

ServiGo uses both MySQL, Firebase Firestore and Firebase Storage to manage data. MySQL database is used to manage booking data, manage customer review data and admin data. Firebase Firestore is used to manage users data (technicians and customers data), service data and technician location data. Friebase storage is used to manage users profile pictures and image slider pictures. The URLs of users profile picture are storing in users collection in firebase firestore.

5.4. API Endpoints (PHP Backend)

Endpoint	Method	Description
saveBooking.php	POST	Saving new bookings after customer confirm it with payment.
getBookings.php	GET	Fetch bookings for display it in customer's booking history.
getTechnicianBookings.php	GET	Retrieve technician assigned bookings for display it in technician's booking history.
updateBookingStatus.php	POST	Update booking status by admin and technician.
submitReview.php	POST	Save user feedback after submit a review.
getReviews.php	GET	Retrieve reviews for a service to display it in single service details screen.

Table 3: API Endpoints

5.5. Security Measures

Several measures are used in ServiGo to protect users, booking and service data.

- ServiGo managing user sign in and sign up process with Firebase Authentication.
- ServiGo is developed using role based accessing. Only customer allows to see promotion banners, image sliders and popular services list. That ensures authorization.
- All the inputs of ServiGo is validated and it prevent the entering invalid and malicious data to the system.
- ServiGo sharing data between client and server through secure API.
- Admin allows to manage users like ban unban from the system securely.

5.6. Deployment

ServiGo is deployed with multiple platforms

- Android Application: Installed on user device as packaged APK
- Backend API: Hosted on web server with apache
- Database: MySQL server configured
- Firebase Services: Configured for authentication, storage, real time database, push notifications.

6. Result and Evaluation

ServiGo system was developed successfully and tested to analyse system performance, functionality, efficiency and reliability. The goal of evaluation is to ensure the system meets the requirements that mentioned in earlier stage.

- Unit testing: This app used unit test to ensure individual components such as API endpoints, android modules are working as expected.
- Integration testing: Integration testing used to ensure good communication between API, mobile app and databases.
- System Testing: The complete work flow of the ServiGo is verified by using system testing.
- User Acceptance testing: The system is with real group of users, and that users verified the system meets business requirements and ready for real world production.

6.1. System Functionality testing

The main feature of the ServiGo such as service booking, payment processing, real-time tracking, and admin management were tested. It ensures ServiGo system allows:

- Users to sign in and sign up successfully without any confusion and error by using Firebase Authentication.
- Customer to search services and schedule a booking after get an idea from other customers reviews without any errors.
- Admin to see that the bookings are listing successfully without error by retrieving data from the backend database.
- Customer to make payment to confirm booking through the payhere without physical cash securely. The bookings are stored only after the confirmation payment.

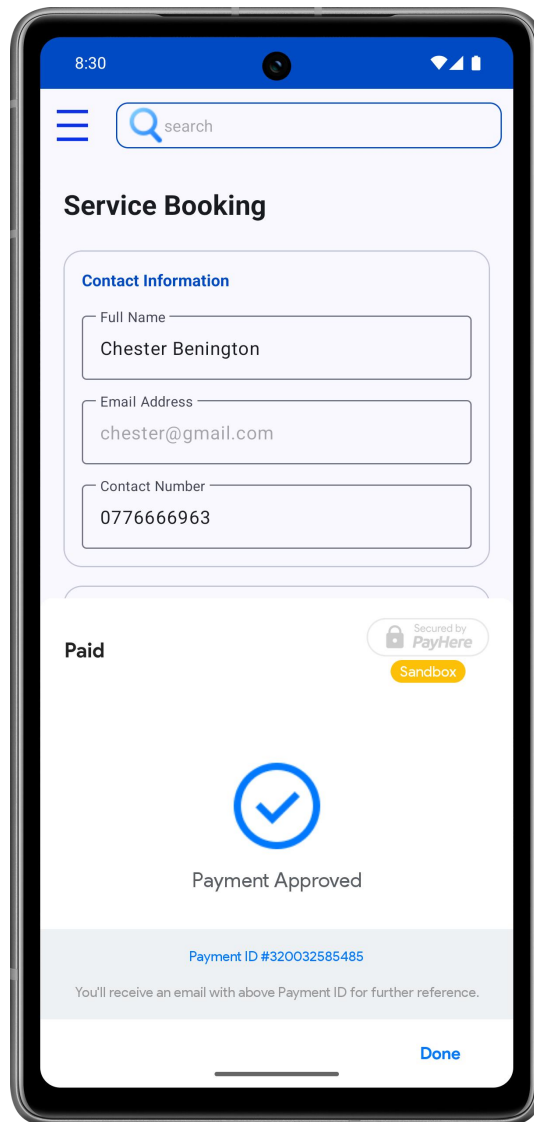


Figure 38: Payment approved

- Customer to track technician in real time through Google map accurately.
- Admin functionalities such as assigning technicians, updating booking status, and managing users worked as expected.

6.2. System performance

API response and requests are performing successfully as expected. And the firebase real time data updates ensure minimal delay. Push Notifications are also performed as expected.

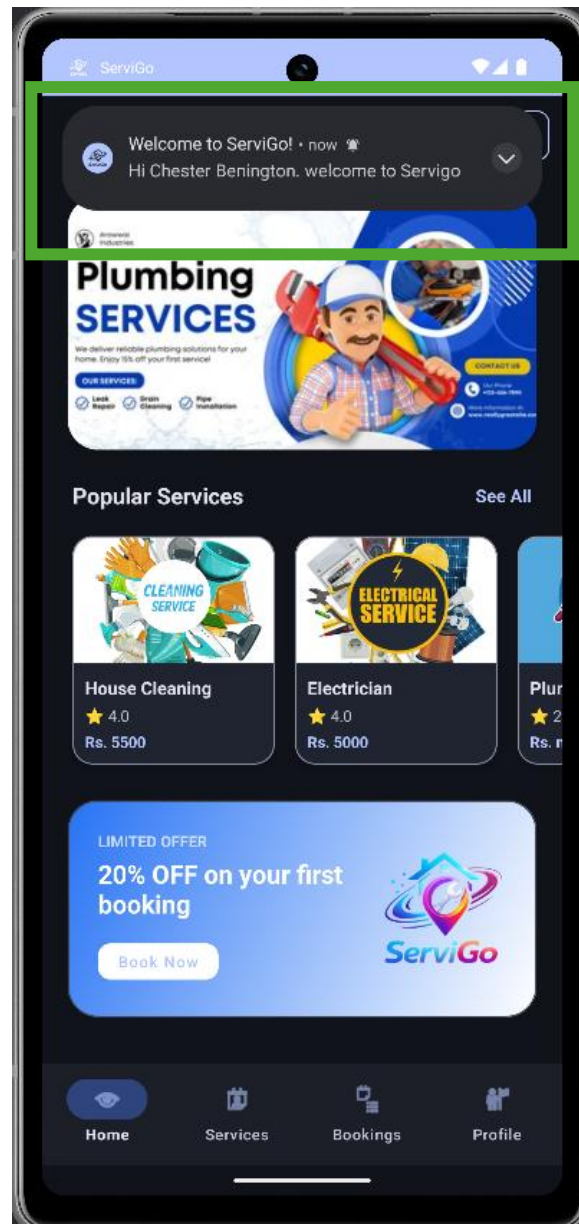


Figure 39: Push Notification

6.3. User Experience Evaluation

The ServiGo system is tested by a group of users.

- User found the simple user friendly interface after login and user navigate between the screens easily without any confusion.
- Some users search services on search bar, some users navigate to services screen and viewed the service details in single service details screen.

- Users got a ideas about services from ratings and reviews before schedule a booking.
- Real time updates on firebase is help to improve users trust, confidence and transparency of the service.

6.4. Limitaions

- The system mainly depends on the internet. Because for the system is runs based on real time data.
- ServiGo only provide single payment gateway.
- Process of technician assignment is done manually by the administrators. It is not good for large scale usage.

6.5. Overall Evaluation

The ServiGo system is meets the requirement and outputs as expected. This system uses modern technologies such as firebase real time updates, mobile payments, real time tracking. That ensures this system is provide complete solution for service booking, real time tracking and simplify the users tasks.

7. Future Work

- Add more payment gateway for more flexibility
- Implement multi language support
- Current application allows get calls between technicians and customers. But there is no chatting channel. Crate chatting feature with firebase will be more useful in the future.
- Implement AI, for get AI based recommendations
- Add a dashboard to view earnings of technicians.
- Advanced reporting features for administrator
- Automatically technician assignment

8. Conclusion

ServiGo project started with two goals. To show Sri Lanka home service market is ready for a new step with ServiGo mobile application. And This project prove a one student of a university could develop the system within a university project.

Both goals are turn out to be true but it was not easy. The market gap is clearly real because there is no platform for home service marketplace in Sri Lanka. That is a problem worth solving. ServiGo fills that both technical gap and practical gap. On technical side, ServiGo is fully build and tested. The app uses firebase , Google map, online payments, live location

tracking and push notification, motion sensors. Also uses web site for the admin panel. These works together smoothly as one experience.

The main highlight of the ServiGo is live technician tracking screen. Users able to see the technician marker and route on map and moving it with technician in real time. Additionally uses scalable technology such as RESTful APIs that ensures system can manage future improvements.

9. Reflection

The development of the ServiGo project is more valuable learning experience. It improved my problem solving skills and the technical skills. Through this project I got experience in developing mobile app using backend services and web based admin panel.

In this project I was understanding how to work with different technologies for a one project. I was learned to connect android mobile app with Firebase authentication, PHP base REST API and implement a Laravel website for admin panel. This helps to understand the architecture of full stack development.

Additionally improved my skill to design user friendly, simple and responsive interfaces. Development of real time tracking and online payments are lead me to improve my logical thinking and manage work flow.

I also learned how to debug errors, manage performance, make reliable and efficient application. This project improve my skill to adapt to the challenges and analytical thinking.

10. References

- Google (n.d.). Google Maps Documentation. Available at: <https://developers.google.com/maps> (Accessed: 2026).
- Firebase (n.d.). Firebase Documentation. Available at: <https://firebase.google.com/docs> (Accessed: 2026).
- PayHere (n.d.). PayHere Developer Documentation. Available at: <https://support.payhere.lk/api> (Accessed: 2026).
- Android Studio (n.d.). Android Developer Documentation. Available at: <https://developer.android.com> (Accessed: 2026).
- Laravel (n.d.). Laravel Official Documentation. Available at: <https://laravel.com/docs> (Accessed: 2026).
- MySQL (n.d.). MySQL Documentation. Available at: <https://dev.mysql.com/doc> (Accessed: 2026).
- phpMyAdmin (n.d.). phpMyAdmin Documentation. Available at: <https://www.phpmyadmin.net/docs> (Accessed: 2026).
- Retrofit (n.d.). Retrofit Documentation. Available at: <https://square.github.io/retrofit/> (Accessed: 2026).